

MHD Derivations for stellar2d — a sympy-verified manuscript

stellar2d development notes
Department of Astronomy, Tsinghua University
github.com/MahoMaho-Rize/stellar2d

2026-05-08

Contents

1	Front matter	4
1.1	Purpose	4
1.2	Reproducibility protocol	5
1.3	Conventions	5
1.4	How to regenerate this manuscript	5
2	A1. Ideal MHD equations (conservative form)	5
2.1	Starting assumptions	6
2.2	Mass conservation	6
2.3	Lorentz force identity (A1-lorentz)	6
2.4	Momentum conservation (divergence form)	6
2.5	Induction equation — two equivalent forms (A1-induction)	7
2.6	Total-energy conservation (A1-energy)	7
2.7	Solenoidal constraint preservation (A1-divB)	7
2.8	Compact conservative system	8
2.9	□ Verification checkpoint (to be wired)	8
3	A2. Conservative $\leftrightarrow$ primitive variable transformation	8
3.1	Variable inventory	8
3.2	Pressure extraction (primitive from conservative)	9
3.3	Forward Jacobian $\partial\mathbf{U}/\partial\mathbf{W}$	9
3.4	Backward Jacobian $\partial\mathbf{W}/\partial\mathbf{U}$	9
3.5	Consistency check (sympy)	9
3.6	□ Verification checkpoint (to be wired)	10
4	A3. Flux Jacobian $\mathbf{A} = \partial\mathbf{F}/\partial\mathbf{U}$ and its 7-wave eigensystem	10
4.1	Working frame	10
4.2	Primitive-form Jacobian $\mathbf{A}_W$	10
4.3	Characteristic speeds	11
4.4	The seven wave speeds	11
4.5	Numerical verification of the spectrum	11
4.6	Closed-form right-eigenvectors	12
4.7	Degenerate-limit eigenvectors	13
4.8	□ Verification checkpoint (to be wired)	13
5	A4. HLLD intermediate states (Miyoshi-Kusano 2005)	13

5.1	Wave fan	13
5.2	Contact speed (MK Eq. 38)	13
5.3	Star-region state	13
5.4	Critical implementation note	14
5.5	□ Verification	14
6	A5. Constrained Transport and discrete $\nabla \cdot \mathbf{B} = 0$	14
6.1	Grid layout	14
6.2	CT update (Evans-Hawley 1988)	14
6.3	The central identity	14
6.4	Gardiner-Stone 2005 corner-EMF averaging	14
6.5	□ Verification	15
7	A6. Reconstruction (PLM/PPM) and TVD slope limiters	15
7.1	Why this section matters	15
7.2	Definitions	15
7.3	Sweby TVD region (A6-Sweby)	15
7.4	PLM reconstruction and order	16
7.5	PPM (Colella-Woodward 1984) 4-point interpolant	16
7.6	Practical recommendation (Stone+08 Appendix A)	16
7.7	□ Verification checkpoints	16
8	A7. Van-Leer 2 predictor-corrector (Stone-Gardiner 2009)	16
8.1	The scheme	17
8.2	Why VL2 over Strang or CTU	17
8.3	Fourier (von Neumann) analysis on linear advection	17
8.4	Truncation error (A7-consistency)	17
8.5	CFL constraint (1D)	18
8.6	Implementation tips (for <code>athena_mhd_solver.cu</code>)	18
8.7	□ Verification checkpoints	18
9	A8. MHD CFL time-step constraints	18
9.1	Hyperbolic CFL (fast wave dominant)	18
9.2	Parabolic CFL (Ohmic + ambipolar diffusion)	18
9.3	Fast-speed behaviour at degenerate limits (A8-cf-limits)	19
9.4	Combined rule	19
9.5	Kernel-form reduction	19
9.6	Super-time-stepping (optional, future)	19
9.7	□ Verification checkpoints	19
10	A9. HLLD degenerate branches	20
10.1	Three degeneracies of the generic HLLD	20
10.2	D1 — Zero longitudinal field ($B_x = 0$)	20
10.3	D2 — Alfvén locus removable singularity	20
10.4	D3 — $S_L \approx S_R$ (vacuum / strongly-aligned)	21
10.5	Implementation order (most to least specific)	21
10.6	Why this matters	21
10.7	□ Verification checkpoints	21
11	A10. Powell 8-wave source vs Constrained Transport	21
11.1	The 8-wave system (Powell+99)	21
11.2	The non-trivial claim	22
11.3	GLM-MHD contrast (Dedner+02)	22
11.4	Implementation note	22
11.5	□ Verification checkpoints	22

12A11. Linear MHD wave initial conditions (Stone+08 Tab 1)	23
12.1 Why this is the “gold-standard” convergence test	23
12.2 Background (Stone+08 Table 1)	23
12.3 Perturbation	23
12.4 Periods on an L -periodic box	23
12.5 Entropy-mode exception	23
12.6 $\nabla \cdot \mathbf{B} = 0$ at IC (automatic)	24
12.7 Expected error	24
12.8 Practical test protocol	24
12.9 $\square$ Verification checkpoint	24
12.10 Extension: 3D linear wave	24
13B1. Super-radial flux-tube reduction	24
13.1 Geometry	24
13.2 1D flux-tube equations	25
13.3 Flux conservation	25
13.4 MHSE	25
13.5 WKB amplitude (see B2)	25
13.6 $\square$ Verification	25
14B2. WKB wave action conservation for Alfvén waves	25
14.1 Linearised transverse MHD	25
14.2 Elsässer variables	26
14.3 Wave action conservation	26
14.4 $\square$ Verification	26
15B3. Parker critical point on a super-radial flux tube	26
15.1 Parker wind equation	26
15.2 Critical (sonic) point	27
15.3 Asymptotic velocity	27
15.4 $\square$ Verification	27
16B4. Well-balanced MHSE at the operator level	27
16.1 The problem	27
16.2 Well-balanced residual (Bermúdez-Vázquez 1994 / Botta+04)	27
16.3 Preservation of linear-wave dynamics	27
16.4 What MHSE means in a super-radial flux tube	28
16.5 Practical recipe	28
16.6 Failure modes observed elsewhere	28
16.7 Snapshot invalidation	28
16.8 $\square$ Verification checkpoints	28
17C1. Ohmic (resistive) dissipation	28
17.1 Ohm’s law with finite conductivity	29
17.2 Induction equation	29
17.3 Joule heating	29
17.4 CFL for explicit diffusion	29
17.5 $\square$ Verification	29
18C2. Ambipolar diffusion (ion-neutral drift)	29
18.1 Motivation	29
18.2 Parallel / perpendicular current selectivity	30
18.3 Kernel-friendly form	30
18.4 Induction equation	30
18.5 Ambipolar heating	30

18.6 CFL	30
18.7 □ Verification	30
19 C3. Total-energy equation with non-ideal MHD	31
19.1 Total-energy equation	31
19.2 Internal-energy source	31
19.3 Sign convention	31
19.4 Kernel structure	31
19.5 □ Verification	31
20 C4. Saha closure for η_O and η_A	32
20.1 Saha equation (pure hydrogen, LTE)	32
20.2 Diffusivity closures (Draine 1983 / Choi+09)	32
20.3 Weakly-ionised limit ($x_e \ll 1$)	32
20.4 Magnetic Reynolds numbers	32
20.5 Kernel implementation	32
20.6 □ Verification	33
21 C5. Sub-grid turbulent heating closure (Suzuki-Inutsuka 2005)	33
21.1 Why this closure exists	33
21.2 Suzuki-Inutsuka 2005 form	33
21.3 Shoda-Yokoyama 2016 (Elsässer) form	33
21.4 Energy-equation coupling	33
21.5 Timescale bound (explicit-source CFL)	34
21.6 Shimizu+22 scaling	34
21.7 Extraction of $\delta v_\perp$	34
21.8 □ Verification checkpoints	34
22 D1. Ideal MHD in cylindrical (R, ϕ, z) coordinates	34
22.1 Cylindrical operators	35
22.2 Radial momentum (axisymmetric)	35
22.3 Axisymmetric radial-flux conservation	35
22.4 □ Verification	35
23 D2. Shearing-sheet approximation and shearing-periodic BC	35
23.1 Hill expansion	35
23.2 Frame-rotation sources	36
23.3 Shearing-periodic BC	36
23.4 Effective potential is anti-confining	36
23.5 □ Verification	36
24 D3. MRI stress decomposition $\alpha_{SS} = \alpha_R + \alpha_M$	36
24.1 Shakura-Sunyaev stress	36
24.2 Suzuki 2023 sign-quadrant decomposition	37
24.3 Balbus-Hawley MRI dispersion	37
24.4 Parseval for Maxwell	37
24.5 □ Verification	37

1 Front matter

1.1 Purpose

This manuscript is an **internal, reproducibility-grade derivation document** for the MHD extension of stellar2d. It covers four parts:

- **Part A** — ideal MHD equations, conservative / primitive forms, flux Jacobian eigensystem, HLLD intermediate states, constrained transport (CT) preservation of $\nabla \cdot \mathbf{B} = 0$; PLM/PPM reconstruction + TVD slope limiters; VL2 predictor-corrector and its CFL bound (hyperbolic + parabolic); HLLD degeneracy branches; Powell-source vs CT comparison; linear-wave convergence IC.
- **Part B** — reduction to a 1D super-radial flux-tube geometry used by the Suzuki-group stellar-wind codes, including the WKB Alfvén-wave action integral, the Parker critical-point condition in this geometry, and the well-balanced MHSE operator needed to keep the atmosphere quiet for long-time wind runs.
- **Part C** — non-ideal dissipation: Ohmic, ambipolar, and their contributions to the energy equation; closure of the two diffusivities through Saha ionisation; sub-grid turbulent heating closure (Suzuki-Inutsuka 2005) for 1D flux-tube wind runs.
- **Part D** — cylindrical shearing-box MHD, the shearing-periodic boundary condition, and the exact Maxwell-Reynolds stress decomposition of the α_{SS} metric.

1.2 Reproducibility protocol

Every algebraic identity in this document is **mechanically verified by sympy**. Each section corresponds to a script in `docs/mhd_derivations/scripts/<section>.py` that ends with `assert_zero(LHS - RHS, ...)` calls. If any identity cannot be simplified by sympy but is physically correct, we document the alternate verification route (e.g., “manually checked with Wolfram, reason: polynomial simplification blows up”).

The scripts are intentionally **independent**: each one imports only `_common.py` and re-derives everything it needs from first principles. No cross-script caching of intermediate symbolic results. This makes any section independently re-runnable — a prerequisite for trusting the manuscript as a source of truth for the solver implementation.

1.3 Conventions

- Units are Gaussian with 4π absorbed into $\mathbf{B}$, so that the Alfvén speed is simply $c_A = B/\sqrt{\rho}$, matching Stone & Gardiner 2008, Athena++, PLUTO.
- γ always denotes the ratio of specific heats (not the relativistic Lorentz factor).
- Einstein summation is *not* used; sums are written explicitly.
- sympy variable names mirror the mathematical symbols ($\rho = \rho$, $B_x = B_x$, etc.); see `scripts/_common.py` for the full inventory.

1.4 How to regenerate this manuscript

```
cd docs/mhd_derivations
bash run_all.sh           # (re-)runs every scripts/*.py, refreshes output/
bash build_manuscript.sh  # concatenates sections/*.md → manuscript.{md,pdf}
```

If a sympy assertion fails during `run_all.sh`, the build halts and the offending section is flagged. No partial manuscript is emitted.

2 A1. Ideal MHD equations (conservative form)

sympy script: `scripts/a1_ideal_mhd_equations.py` **generated LaTeX:** `output/a1_ideal_mhd_equations.latex.tex` **verifies:** 8 identities (Lorentz $\times$ 3 components, induction $\times$ 3, Poynting-flux identity, $\nabla \cdot (\nabla \times \cdot) = 0$) **code checkpoints (future):** `src/gpu/explicit/athena_mhd_kernels.cu::d_mhd_flux`

2.1 Starting assumptions

Label	Assumption
A1a	Compressible neutral gas with magnetic field $\mathbf{B}$; no viscosity, no Ohmic, no ambipolar, no Hall.
A1b	Infinite conductivity $\rightarrow$ frozen-in flux, $\mathbf{E} = -\mathbf{v} \times \mathbf{B}$.
A1c	Non-relativistic: displacement current $\partial_t \mathbf{E}/c$ dropped.
A1d	Ideal EOS: $p = (\gamma - 1)\rho e$.

2.2 Mass conservation

Written compactly as

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0. \quad (\text{A1-mass})$$

Under A1a-d this is the unmodified fluid continuity equation.

2.3 Lorentz force identity (A1-lorentz)

The single most important vector identity of the derivation is

$$(\nabla \times \mathbf{B}) \times \mathbf{B} = \nabla \cdot (\mathbf{B} \otimes \mathbf{B}) - \nabla \left(\frac{1}{2} |\mathbf{B}|^2 \right) - (\nabla \cdot \mathbf{B}) \mathbf{B}. \quad (\text{A1-lorentz})$$

The last term vanishes **analytically** when the solenoidal constraint $\nabla \cdot \mathbf{B} = 0$ is enforced. At the discrete level it does not vanish in general; this is precisely what CT schemes (§A5) are engineered to handle.

Sympy verification. All three vector components of the residual $(\nabla \times \mathbf{B}) \times \mathbf{B} - [\nabla \cdot (\mathbf{B} \otimes \mathbf{B}) - \nabla(\frac{1}{2}|\mathbf{B}|^2) - (\nabla \cdot \mathbf{B})\mathbf{B}]$ reduce to 0 under `sympy.simplify`, without imposing $\nabla \cdot \mathbf{B} = 0$. This confirms the identity is an *algebraic*, not a physical, fact.

2.4 Momentum conservation (divergence form)

Define the **total magneto-fluid pressure**

$$P^* \equiv p + \frac{1}{2} |\mathbf{B}|^2.$$

Then (A1-lorentz) lets us write momentum conservation in divergence form:

$$\boxed{\partial_t(\rho \mathbf{v}) + \nabla \cdot [\rho \mathbf{v} \otimes \mathbf{v} - \mathbf{B} \otimes \mathbf{B} + P^* \mathbf{I}] = \mathbf{0}.} \quad (\text{A1-mom})$$

Implementation note. $\mathbf{B} \otimes \mathbf{B}$ is *not* symmetric in its indices when read as a flux tensor F_{ij} (it is actually symmetric, but in contrast the induction tensor $v_i B_j - v_j B_i$ is anti-symmetric). In the kernel, $B_i B_j$ for each $(i, j) \in \{(x, x), (x, y), (x, z), (y, y), (y, z), (z, z)\}$ must be computed on the same stencil as $v_i v_j$ — mixing face-averaged $\mathbf{B}$ with cell-centred $\mathbf{B}$ here is the number-one source of initial HLLD bugs.

2.5 Induction equation — two equivalent forms (A1-induction)

Tensor-divergence form (preferred for a Godunov kernel):

$$\partial_t B_i + \partial_j (v_j B_i - v_i B_j) = 0.$$

Curl form (preferred for a CT update):

$$\partial_t \mathbf{B} = \nabla \times (\mathbf{v} \times \mathbf{B}).$$

Sympy verification. For each component $i \in \{x, y, z\}$ we check that

$$\nabla \times (\mathbf{v} \times \mathbf{B}) \Big|_i \equiv -\partial_j (v_j B_i - v_i B_j)$$

holds as a *pure vector-calculus identity* (no $\nabla \cdot \mathbf{B} = 0$, no $\nabla \cdot \mathbf{v} = 0$ invoked). This is the well-known Jacobi-like identity; all three sympy assertions pass.

2.6 Total-energy conservation (A1-energy)

With

$$E \equiv \rho e + \frac{1}{2} \rho |\mathbf{v}|^2 + \frac{1}{2} |\mathbf{B}|^2,$$

the conservative form is

$$\partial_t E + \nabla \cdot [(E + P^*) \mathbf{v} - \mathbf{B}(\mathbf{B} \cdot \mathbf{v})] = 0.$$

The magnetic-Poynting-flux piece $-\mathbf{B}(\mathbf{B} \cdot \mathbf{v})$ and the enthalpy-like piece $P^* \mathbf{v}$ together make the flux manifestly symmetric-hyperbolic.

Sympy-verified identity. The vector identity $\mathbf{B} \times (\mathbf{v} \times \mathbf{B}) = |\mathbf{B}|^2 \mathbf{v} - \mathbf{B}(\mathbf{v} \cdot \mathbf{B})$ lets us rewrite the Poynting flux $\mathbf{S} = \frac{1}{2} \mathbf{B} \times (\mathbf{v} \times \mathbf{B})$ in the form used in (A1-energy). `assert_zero` confirms $\nabla \cdot [\mathbf{B} \times (\mathbf{v} \times \mathbf{B}) - (|\mathbf{B}|^2 \mathbf{v} - \mathbf{B}(\mathbf{B} \cdot \mathbf{v}))] = 0$ identically.

2.7 Solenoidal constraint preservation (A1-divB)

Taking the divergence of the induction equation in curl form:

$$\partial_t (\nabla \cdot \mathbf{B}) = \nabla \cdot \nabla \times (\mathbf{v} \times \mathbf{B}) = 0,$$

where the last equality is the identity $\nabla \cdot (\nabla \times \cdot) = 0$. **At the continuous level this is automatic**; at the discrete level it is what CT or Dedner-GLM machinery must engineer. See §A5.

Sympy verification. `assert_zero(div_cart(curl_cart(vxB)))` passes for arbitrary smooth $\mathbf{v}, \mathbf{B}$.

2.8 Compact conservative system

Collecting all fluxes, we write the 8-variable conservative MHD system:

$$\partial_t \mathbf{U} + \partial_i \mathbf{F}_i(\mathbf{U}) = \mathbf{0}, \quad \mathbf{U} = (\rho, \rho \mathbf{v}, \mathbf{B}, E)^T.$$

The explicit form of $\mathbf{F}_i$ follows by assembling the four fluxes above. We defer the component-by-component flux Jacobian and its eigensystem to §A3.

2.9 □ Verification checkpoint (to be wired)

Implementation invariants the kernel must satisfy (enforced through `tests/test_athena_mhd_*.cu` in a future PR):

1. **One-cell linearised update reduces to (A1-mass), (A1-mom), (A1-induction-tensor), (A1-energy) exactly** — zero the $\mathbf{B}$ field and verify that $\mathbf{F}_i$ reduces to the Euler flux used in `athena_vl2`.
2. **Solenoidal residual** $(\nabla \cdot \mathbf{B})_{cell}$ from face-centred $\mathbf{B}$ remains at round-off over 100 sound-crossing times on a smooth periodic IC (field-loop advection, §A5).
3. **Energy-momentum consistency** — on a piecewise-constant linear wave IC, total energy updated via (A1-energy) agrees with $\frac{1}{2}\rho|\mathbf{v}|^2 + \frac{1}{2}|\mathbf{B}|^2 + \rho e$ reconstructed from the primitive fields, to round-off per step.

Any failure on (1)–(3) flags the implementation as inconsistent with §A1 and must be resolved before proceeding to §A2.

3 A2. Conservative $\leftrightarrow$ primitive variable transformation

sympy script: `scripts/a2_conservative_primitive.py` **generated LaTeX:** `output/a2_conservative_primitive.latex.tex` **verifies:** $\frac{\partial \mathbf{W}}{\partial \mathbf{U}} \cdot \frac{\partial \mathbf{U}}{\partial \mathbf{W}} = \mathbf{I}_8$ (all 64 entries symbolically checked to zero after subtraction). **code checkpoints (future):** `athena_mhd_kernels.cu::d_prim_from_cons, d_cons_from_prim`.

3.1 Variable inventory

The conservative 8-vector used inside the Godunov flux loop:

$$\mathbf{U} = (\rho, \rho v_x, \rho v_y, \rho v_z, B_x, B_y, B_z, E)^T.$$

The primitive 8-vector used for reconstruction / Riemann solver input:

$$\mathbf{W} = (\rho, v_x, v_y, v_z, B_x, B_y, B_z, p)^T.$$

The total-energy density closes the map through the ideal EOS:

$$E = \frac{p}{\gamma - 1} + \frac{1}{2}\rho|\mathbf{v}|^2 + \frac{1}{2}|\mathbf{B}|^2. \quad (\text{A2-E})$$

3.2 Pressure extraction (primitive from conservative)

Inverting (A2-E) for p gives the pressure-extraction formula the `d_prim_from_cons` kernel must use:

$$p = (\gamma - 1) \left(E - \frac{|\mathbf{m}|^2}{2\rho} - \frac{1}{2}|\mathbf{B}|^2 \right), \quad \mathbf{m} \equiv \rho\mathbf{v}.$$

Implementation gotcha. For low- β , high-Mach flows (MHD blast, Orszag-Tang late time), the hydro term $|\mathbf{m}|^2/(2\rho)$ and the magnetic term $|\mathbf{B}|^2/2$ can both be within machine-precision of the total E . A naïve subtraction can then hand the EOS a negative p . The kernel must either:

1. Accept a positivity fallback (fall back to isothermal primitive extraction and flag the cell),
or
2. Use the dual-energy formulation (evolve an auxiliary internal-energy variable and use it when the primitive subtraction is unreliable).

This is documented in Stone & Gardiner 2008 §4.6 and implemented in Athena++ as the `DUAL_ENERGY` compile-time flag.

3.3 Forward Jacobian $\partial\mathbf{U}/\partial\mathbf{W}$

sympy produces the 8×8 Jacobian explicitly; it is block-sparse with the $\mathbf{B}$ block trivially the identity, the momentum block $\rho\mathbf{I} + \mathbf{v} \otimes (\partial\mathbf{v}/\partial\mathbf{W})$, and the energy row containing $\partial E/\partial\rho$, $\rho\mathbf{v}$, $\mathbf{B}$, $1/(\gamma - 1)$.

3.4 Backward Jacobian $\partial\mathbf{W}/\partial\mathbf{U}$

The inverse map takes the conservative 8-vector to primitives term by term. The non-trivial rows are the velocity-from-momentum and the pressure-from-energy rows:

$$\begin{aligned} \frac{\partial v_i}{\partial(\rho v_j)} &= \frac{\delta_{ij}}{\rho}, & \frac{\partial v_i}{\partial\rho} &= -\frac{v_i}{\rho}, \\ \frac{\partial p}{\partial\rho} &= (\gamma - 1) \frac{|\mathbf{m}|^2}{2\rho^2} = (\gamma - 1) \frac{1}{2}|\mathbf{v}|^2, \\ \frac{\partial p}{\partial(\rho v_i)} &= -(\gamma - 1)v_i, & \frac{\partial p}{\partial B_i} &= -(\gamma - 1)B_i, & \frac{\partial p}{\partial E} &= \gamma - 1. \end{aligned}$$

3.5 Consistency check (sympy)

The script computes $\frac{\partial\mathbf{W}}{\partial\mathbf{U}}(\mathbf{U}(\mathbf{W})) \cdot \frac{\partial\mathbf{U}}{\partial\mathbf{W}}(\mathbf{W})$ after substituting back so that both Jacobians are evaluated at the same primitive state, and verifies

$$\frac{\partial\mathbf{W}}{\partial\mathbf{U}} \cdot \frac{\partial\mathbf{U}}{\partial\mathbf{W}} = \mathbb{I}_8. \quad (\text{A2-inverse})$$

All 64 entries of the residual matrix are `sympy.simplify-reduced` to 0 and pass `assert_zero`.

3.6 □ Verification checkpoint (to be wired)

When the MHD kernel is written, the test

tests/test_athena_mhd_roundtrip.cu

should:

1. Seed random primitive 8-vectors $\mathbf{W}$ with $\rho, p > 0$ and $|\mathbf{B}|^2/(2p) < 10^6$ (stay out of the positivity danger zone).
2. Call `d_cons_from_prim` to compute $\mathbf{U}$.
3. Call `d_prim_from_cons` to recover $\mathbf{W}'$.
4. Assert $\|\mathbf{W} - \mathbf{W}'\|_\infty / \|\mathbf{W}\|_\infty < 10 \varepsilon_{\text{mach}}$.

Any regression of (A2-inverse) in the kernel fails this round-trip.

4 A3. Flux Jacobian $\mathbf{A} = \partial \mathbf{F} / \partial \mathbf{U}$ and its 7-wave eigensystem

sympy script: scripts/a3_flux_jacobian_eigensystem.py **generated LaTeX:** output/a3_flux_jacobian_eigensystem.latex.tex **symbolically verified:** hydro projection (25 entries), discriminant identities I1 & I2. **numerically verified (20 trials $\times$ 3 γ):** eigenvalues match closed-form to $\leq 5 \times 10^{-15}$; $\mathbf{A}\mathbf{r} - \lambda\mathbf{r}$ residuals $\leq 6 \times 10^{-15}$. **code checkpoints (future):** athena_mhd_kernels.cu::d_mhd_eigenvalues_primitive, d_mhd_right_eigenvectors.

4.1 Working frame

The eigenanalysis is done in **primitive form** in 1D along $\hat{x}$. Primitive 7-vector:

$$\mathbf{W} = (\rho, v_x, v_y, v_z, B_y, B_z, p)^T.$$

B_x is **not** an evolution variable in 1D: $\partial_x B_x = \nabla \cdot \mathbf{B} = 0$ forces $B_x = \text{const}$ across any x -interface. The 1D MHD system therefore has **seven** propagating waves, not eight; the eighth mode of the 3D system is the divergence-cleaning wave handled separately by CT (§A5).

4.2 Primitive-form Jacobian $\mathbf{A}_W$

$$\partial_t \mathbf{W} + \mathbf{A}_W(\mathbf{W}) \partial_x \mathbf{W} = \mathbf{0},$$

$$\mathbf{A}_W = \begin{bmatrix} v_x & \rho & 0 & 0 & 0 & 0 & 0 \\ 0 & v_x & 0 & 0 & B_y/\rho & B_z/\rho & 1/\rho \\ 0 & 0 & v_x & 0 & -B_x/\rho & 0 & 0 \\ 0 & 0 & 0 & v_x & 0 & -B_x/\rho & 0 \\ 0 & B_y & -B_x & 0 & v_x & 0 & 0 \\ 0 & B_z & 0 & -B_x & 0 & v_x & 0 \\ 0 & \gamma p & 0 & 0 & 0 & 0 & v_x \end{bmatrix}. \quad (\text{A3-jacobian})$$

Hydrodynamic sanity check. Setting $B_x = B_y = B_z = 0$ and projecting onto the (ρ, v_x, v_y, v_z, p) subspace recovers the standard 5×5 hydrodynamic primitive Jacobian. sympy verifies all 25 entries of the projection match the hydro reference.

4.3 Characteristic speeds

Define the four reference speeds:

$$c_{s_0}^2 \equiv \gamma p / \rho, \quad c_{Ax}^2 \equiv B_x^2 / \rho, \quad c_{A\perp}^2 \equiv (B_y^2 + B_z^2) / \rho, \quad c_A^2 \equiv c_{Ax}^2 + c_{A\perp}^2.$$

The fast and slow magnetosonic speeds are the two positive roots of

$$c_{f,s}^2 = \frac{1}{2} \left[(c_{s_0}^2 + c_A^2) \pm \sqrt{(c_{s_0}^2 + c_A^2)^2 - 4c_{s_0}^2 c_{Ax}^2} \right]. \quad (\text{A3-cfs})$$

Discriminant identities (HLLD-stable forms). The closed form (A3-cfs) loses ULPs when $c_{A\perp} \rightarrow 0$ or $c_{s_0} \rightarrow 0$. The HLLD kernel uses the two equivalent identities

$$\boxed{c_f^2 + c_s^2 = c_{s_0}^2 + c_A^2, \quad c_f^2 \cdot c_s^2 = c_{s_0}^2 \cdot c_{Ax}^2}, \quad (\text{A3-discriminant})$$

which sympy verifies symbolically (both reduce to 0 under simplify). With these two identities, c_f and c_s can be recovered from $(c_{s_0}^2, c_A^2, c_{Ax}^2)$ without ever forming the root-of-difference $c_A^2 - c_{s_0}^2$.

4.4 The seven wave speeds

$$\boxed{\{\lambda_k\}_{k=1}^7 = \{v_x - c_f, v_x - c_{Ax}, v_x - c_s, v_x, v_x + c_s, v_x + c_{Ax}, v_x + c_f\}}. \quad (\text{A3-wave-speeds})$$

Here $c_{Ax} \equiv B_x / \sqrt{\rho}$ carries the sign of B_x — this convention lets us write the Alfvén speed once without the $s = \text{sign}(B_x)$ factor that appears in Stone+08 eigenvector formulas. The other convention (unsigned $c_{Ax} = |B_x| / \sqrt{\rho} + s$ in every eigenvector entry) is equivalent; we choose signed- c_{Ax} here because it keeps the 7-wave spectrum contiguous for all $\text{sign}(B_x)$.

4.5 Numerical verification of the spectrum

sympy’s `simplify()` does not handle nested radicals $\sqrt{a \pm \sqrt{b}}$ reliably; the fast / slow eigenvector residuals cannot be reduced to zero purely symbolically. This is a known limit — Stone+08 Appendix B, Roe & Balsara 1996, and the Athena++ source all fall back to **numerical random-sample verification** (the “Roe check”) for the eigensystem. We do the same:

1. Draw 20 random physically admissible states $(\rho, p, v_x, B_x, B_y, B_z)$ with $\rho, p > 0$, $B_x \neq 0$, $|B_\perp| \neq 0$;
2. For each of three ratios of specific heats $\gamma \in \{5/3, 7/5, 4/3\}$, diagonalise A_W numerically with `numpy.linalg.eig`;
3. Sort the seven numerical λ_k and compare to the sorted closed-form spectrum (A3-wave-speeds).

Results over 60 trials:

$$\max |\lambda_{\text{closed}} - \lambda_{\text{numerical}}| < 5 \times 10^{-15}, \quad \max \|\mathbf{A}\mathbf{r} - \lambda\mathbf{r}\|_\infty < 6 \times 10^{-15}.$$

The residuals are at the double-precision floor for matrix eigenvalue computation on a well-conditioned 7×7 matrix — indistinguishable from exact.

4.6 Closed-form right-eigenvectors

The closed forms come from Stone & Gardiner 2008 Appendix B Eqs. (B11)–(B13); they are reproduced verbatim in the Athena++ source (`src/eos/adiabatic_mhd.cpp::LRMHDWaves`). We document them here for the kernel implementation; their correctness is pinned to the community verification in Stone+08 Fig. 28–30.

Define the **amplitude-normalisation coefficients**

$$\alpha_f^2 = \frac{c_{s_0}^2 - c_s^2}{c_f^2 - c_s^2}, \quad \alpha_s^2 = \frac{c_f^2 - c_{s_0}^2}{c_f^2 - c_s^2}, \quad \alpha_f^2 + \alpha_s^2 = 1,$$

and the **direction coefficients**

$$\beta_y = \frac{B_y}{\sqrt{B_y^2 + B_z^2}}, \quad \beta_z = \frac{B_z}{\sqrt{B_y^2 + B_z^2}}, \quad \beta_y^2 + \beta_z^2 = 1,$$

and $s = \text{sign}(B_x)$.

4.6.1 Entropy (contact) wave — $\lambda_4 = v_x$

$$\mathbf{r}_{\text{entropy}} = (1, 0, 0, 0, 0, 0, 0)^T.$$

Pure density perturbation, no velocity, pressure, or magnetic-field change. This is literally what “contact discontinuity” means in MHD — it propagates unchanged at the flow speed.

4.6.2 Alfvén waves — $\lambda_{2,6} = v_x \mp c_{Ax}$

$$\mathbf{r}_{A\pm} = (0, 0, \mp s\beta_z, \pm s\beta_y, -\beta_z\sqrt{\rho}, \beta_y\sqrt{\rho}, 0)^T.$$

Transverse velocity and B -field perturbation, no density or pressure change. This is the canonical transverse electromagnetic wave of MHD.

4.6.3 Fast magnetosonic waves — $\lambda_{1,7} = v_x \mp c_f$

$$\mathbf{r}_{f\pm} = (\rho\alpha_f, \pm\alpha_f c_f, \mp s\alpha_s c_s \beta_y, \mp s\alpha_s c_s \beta_z, \alpha_s c_{s_0} \sqrt{\rho} \beta_y, \alpha_s c_{s_0} \sqrt{\rho} \beta_z, \alpha_f \gamma p)^T.$$

Longitudinal compression plus in-phase perpendicular- B compression.

4.6.4 Slow magnetosonic waves — $\lambda_{3,5} = v_x \mp c_s$

$$\mathbf{r}_{s\pm} = (\rho\alpha_s, \pm\alpha_s c_s, \pm s\alpha_f c_f \beta_y, \pm s\alpha_f c_f \beta_z, -\alpha_f c_{s_0} \sqrt{\rho} \beta_y, -\alpha_f c_{s_0} \sqrt{\rho} \beta_z, \alpha_s \gamma p)^T.$$

Longitudinal compression plus out-of-phase perpendicular- B compression.

4.7 Degenerate-limit eigenvectors

When $|\mathbf{B}_\perp| = 0$ (pure longitudinal $\mathbf{B}$), the β_y, β_z definitions above divide by zero. Stone+08 Eq. (B17)-(B20) gives replacement eigenvectors for this case. We do not verify these here — they are documented in the Athena++ source and must be implemented as branch-conditionals in the kernel.

4.8 $\square$ Verification checkpoint (to be wired)

The future

tests/test_athena_mhd_eigensystem.cu

must:

1. For 100 random admissible states, evaluate `d_mhd_eigenvalues_primitive` and verify each returned λ_k agrees with (A3-wave-speeds) to 10^{-12} absolute.
2. For each state, evaluate `d_mhd_right_eigenvectors` and verify $\|A_W(\mathbf{W}) \mathbf{r}_k - \lambda_k \mathbf{r}_k\|_\infty < 10^{-12}$ for all $k \in \{1, \dots, 7\}$.
3. Pressure-positivity stress test: states with $\beta_{\text{plasma}} < 0.01$, verify no NaN / no overflow in α_f, α_s .
4. Degenerate-limit coverage: states with $|\mathbf{B}_\perp|/|\mathbf{B}| < 10^{-10}$, verify the degenerate-limit branch is taken and produces non-zero eigenvectors.

This eigenanalysis is the algebraic backbone of HLLD (§A4); any regression here is a direct bug in every Riemann flux computation.

5 A4. HLLD intermediate states (Miyoshi-Kusano 2005)

sympy script: `scripts/a4_hlld_intermediate_states.py` **verified:** $p_{\text{tot,L}}^* = p_{\text{tot,R}}^*$ symbolically; Rankine-Hugoniot outer-wave jumps numerically (20 trials, max err 3×10^{-15}). **code checkpoints:** `athena_mhd_kernels.cu::d_hlld_flux`.

5.1 Wave fan

HLLD partitions the Riemann fan between S_L and S_R into four intermediate plateaus separated by S_L^*, S_M^*, S_R^* — outer fast, Alfvén, contact, Alfvén, outer fast.

5.2 Contact speed (MK Eq. 38)

$$S_M = \frac{(S_R - v_{xR})\rho_R v_{xR} - (S_L - v_{xL})\rho_L v_{xL} - p_{\text{tot,R}}^* + p_{\text{tot,L}}^*}{(S_R - v_{xR})\rho_R - (S_L - v_{xL})\rho_L}.$$

5.3 Star-region state

$\rho_K^* = \rho_K(S_K - v_{xK})/(S_K - S_M)$ (MK Eq. 43); $p_{\text{tot}}^* = p_{\text{tot,L}}^* + \rho_L(S_L - v_{xL})(S_M - v_{xL}) = p_{\text{tot,R}}^* + \rho_R(S_R - v_{xR})(S_M - v_{xR})$ (MK Eq. 41). **Sympy symbolically verifies** the L/R equality.

B_{yK}^*, v_{yK}^* from MK Eqs. 44-47 (see script). Alfvén speeds $S_{L,R}^* = S_M \mp |B_x|/\sqrt{\rho_{L,R}^*}$.

5.4 Critical implementation note

HLLD star state is NOT an EOS state. Do NOT compute $\mathbf{F}_K^*$ via $p = (\gamma - 1)(E^* - \frac{1}{2}\rho|\mathbf{v}|^2 - \frac{1}{2}|\mathbf{B}|^2)$ — it gives the wrong pressure. Use MK Eq. 64 directly with p_{tot}^* from (A4-Ptot-star). Numerical verification confirms: RH residuals are $\mathcal{O}(1)$ with EOS inversion, $\mathcal{O}(10^{-15})$ with direct MK formula.

5.5 $\square$ Verification

tests/test_athena_mhd_hlld.cu — Brio-Wu shock tube match Stone+08 Fig 28 to $L^1 < 2\%$ at $N = 512$.

6 A5. Constrained Transport and discrete $\nabla \cdot \mathbf{B} = 0$

sympy script: scripts/a5_ct_divergence_preservation.py **verified:** CT telescoping identity $(\nabla \cdot \mathbf{B})^{n+1} - (\nabla \cdot \mathbf{B})^n = 0$; Gardiner-Stone 2005 corner-EMF averaging is 2nd-order accurate. **code checkpoints:** athena_mhd_solver.cu::update_face_B_with_emf, tests/test_athena_mhd_field_loop.cu.

6.1 Grid layout

Yee-like staggered: B_x on vertical faces $(i \pm \frac{1}{2}, j)$, B_y on horizontal faces $(i, j \pm \frac{1}{2})$, E_z on corners.

6.2 CT update (Evans-Hawley 1988)

$$B_x^{i\pm 1/2, j, n+1} = B_x^{i\pm 1/2, j, n} - \frac{\Delta t}{\Delta y} (E_z^{i\pm 1/2, j+1/2} - E_z^{i\pm 1/2, j-1/2}),$$

$$B_y^{i, j\pm 1/2, n+1} = B_y^{i, j\pm 1/2, n} + \frac{\Delta t}{\Delta x} (E_z^{i+1/2, j\pm 1/2} - E_z^{i-1/2, j\pm 1/2}).$$

6.3 The central identity

$$\boxed{(\nabla \cdot \mathbf{B})_{i,j}^{n+1} = (\nabla \cdot \mathbf{B})_{i,j}^n}$$

Sympy symbolically verifies this as a pure telescoping identity — the four corner-EMF contributions cancel exactly, regardless of what values the EMFs take. This is why CT is qualitatively different from Dedner GLM: $\nabla \cdot \mathbf{B} = 0$ is an **exact** property, not a truncation error.

6.4 Gardiner-Stone 2005 corner-EMF averaging

$$E_z^{i+1/2, j+1/2} = \frac{1}{4} (E_z^{x, i+1/2, j} + E_z^{x, i+1/2, j+1} + E_z^{y, i, j+1/2} + E_z^{y, i+1, j+1/2}).$$

Sympy-verified via Taylor expansion: leading error is $\frac{h^2}{8}(\partial_x^2 E_z + \partial_y^2 E_z) + \mathcal{O}(h^4)$, no $\mathcal{O}(h)$ term — matches 2nd-order Godunov accuracy.

6.5 □ Verification

tests/test_athena_mhd_field_loop.cu — GS05 field-loop, 10 crossings, lock max $|\nabla \cdot \mathbf{B}| < 10\epsilon_{\text{mach}}$ at every step.

7 A6. Reconstruction (PLM/PPM) and TVD slope limiters

sympy script: scripts/a6_reconstruction_tvd.py **verified:** minmod / vL / MC opposite-sign cancellation; Sweby TVD region $0 \leq \varphi(r) \leq \min(2, 2r)$ for $r \in [0, 4]$; PLM reconstruction second-order; PPM 4-point interpolant 4th-order (exact to cubics); PPM parabola $\int_0^1 W d\xi = a_0$. **code checkpoints:** athena_mhd_kernels.cu::d_reconstruct_primitive_plm, athena_mhd_kernels.cu::d_reconstruct_primitive_ppm, tests/test_athena_mhd_linear_wave_convergence.cu.

7.1 Why this section matters

VL2 + HLLD without a **monotonic** reconstruction is 2nd-order but gives spurious oscillations at every shock. The three limiters below are the only ones we need in practice: **minmod** (safest, dispersive), **van Leer** (smooth, slightly sharper), **MC** (most aggressive, Stone+08 default).

A skipped step here directly causes the class of bugs already seen on stellar2d:

- cart_ale2 swept-remap used donor-cell then MUSCL without characteristic projection — caused the periodic-BC drift at P30/P31.
- Stone+08 App. A is explicit: **reconstruct in primitive variables, then project onto characteristic variables** via the eigenvectors of §A3. Skipping the projection converts crisp fast-mode jumps into diffusive blobs.

7.2 Definitions

$$\sigma_{\text{minmod}}(a, b) = \begin{cases} 0, & \text{sign}(a) \neq \text{sign}(b), \\ \text{sign}(a) \min(|a|, |b|), & \text{else.} \end{cases}$$

$$\sigma_{\text{vL}}(a, b) = \begin{cases} 0, & ab \leq 0, \\ \frac{2ab}{a+b}, & ab > 0. \end{cases}$$

$$\sigma_{\text{MC}}(a, b) = \text{sign}(a) \min\left(2|a|, \frac{|a+b|}{2}, 2|b|\right) \text{ when } ab > 0, \text{ else } 0.$$

7.3 Sweby TVD region (A6-Sweby)

All three limiters satisfy, with $r = b/a$:

$$0 \leq \varphi(r) \leq \min(2, 2r), \quad r \geq 0.$$

Sympy-verified by numerical sweep ($r \in [0, 4]$, 401 samples). Any limiter in this region is TVD for 1D scalar advection with $\text{CFL} \leq 1$ (Harten 1983; LeVeque 2002 §16).

7.4 PLM reconstruction and order

$$W_{i+1/2}^L = W_i + \frac{1}{2}\sigma_i, \quad \sigma_i = \frac{1}{2}(W_{i+1} - W_{i-1}).$$

Leading error $W_{i+1/2}^L - W(x_{i+1/2}) = -\frac{h^2}{8}W''(x_{i+1/2}) + \mathcal{O}(h^3)$, so the reconstruction is **second order** (no $\mathcal{O}(h)$ term). Sympy-verified: no h^0 or h^1 terms in the residual expansion.

7.5 PPM (Colella-Woodward 1984) 4-point interpolant

$$W_{i+1/2} = \frac{7}{12}(W_i + W_{i+1}) - \frac{1}{12}(W_{i-1} + W_{i+2}),$$

applied to the **cell-averaged** values. Sympy-verified: no $\mathcal{O}(h^0..h^3)$ term in the expansion against the smooth reference; leading error is $-\frac{1}{30}h^4W^{(4)}$.

The parabolic form inside a cell (Colella-Woodward 1984 Eq. 1.6),

$$W(\xi) = a_L + \xi[\Delta a + a_6(1 - \xi)], \quad \Delta a = a_R - a_L, \quad a_6 = 6(a_0 - (a_L + a_R)/2),$$

satisfies $\int_0^1 W d\xi = a_0$ **exactly** (sympy-verified): conservation of cell averages is built in.

7.6 Practical recommendation (Stone+08 Appendix A)

1. Reconstruct $(\rho, \mathbf{v}, \mathbf{B}, p)$ in **primitive** variables (not conservative). This keeps positivity of ρ, p easier to enforce.
2. Project onto characteristic variables $\delta W^{(k)} = \ell_k \cdot \delta W$ using the left- eigenvectors of §A3, limit each wave family separately, then project back $\delta W = \sum_k r_k \delta W^{(k)}$.
3. Use MC as the default; switch to van Leer near strong shocks if MC produces staircase artefacts (rare but documented in Stone+08 Fig 28).

7.7 □ Verification checkpoints

- tests/test_athena_mhd_linear_wave_convergence.cu — 3-resolution linear fast-wave advection; L^1 convergence slope in $[1.9, 2.1]$.
- tests/test_athena_mhd_shock_tube.cu — Brio-Wu, MC limiter, no staircase within the fast rarefaction fan.

8 A7. Van-Leer 2 predictor-corrector (Stone-Gardiner 2009)

sympy script: scripts/a7_vl2_predictor_corrector.py **verified:** amplification factor $g(\xi) = 1 - i\nu\xi - \frac{\nu^2}{2}\xi^2 + \mathcal{O}(\xi^3)$ (Lax-Wendroff 2nd-order); $|g|^2 \leq 1$ on $|\nu| \leq 1$; truncation error $\mathcal{O}(\Delta t^3 + \Delta t \Delta x^2)$. **code checkpoints:** athena_mhd_solver.cu::vl2_predictor_step, athena_mhd_solver.cu::vl2_corrector_step.

8.1 The scheme

$$\mathbf{U}^* = \mathbf{U}^n + \frac{\Delta t}{2} \mathcal{L}[\mathbf{U}^n], \quad \mathbf{U}^{n+1} = \mathbf{U}^n + \Delta t \mathcal{L}[\mathbf{U}^*].$$

This is **midpoint-RK2** in time, paired with any semi-discrete operator $\mathcal{L}$ (here: PLM reconstruction + HLLD flux from §A4, plus CT EMF update from §A5).

8.2 Why VL2 over Strang or CTU

Feature	VL2	Strang	CTU
Dimensional coupling	Unsplit, exact	Split, $\mathcal{O}(\Delta t^2)$ error	Unsplit
Auxiliary state	$\mathbf{U}^*$	per-direction sweep	corner-transport cells
CFL limit (unsplit)	≤ 1	≤ 1 in each direction	≤ 1 (multi-D)
Compatibility with CT	Native (Stone-Gardiner 2009)	Requires extra corner EMF gymnastics	Native
Kernel complexity	Low (2 stages)	Low (per-direction)	Higher (CTU corner sweeps)

Stone+08 CTU is more accurate on discontinuities but significantly more complex; VL2 is the default in Athena++ and what we adopt.

8.3 Fourier (von Neumann) analysis on linear advection

With PLM + upwind flux for $a > 0$ on a smooth Fourier mode,

$$g(\xi) = 1 - i\nu\xi - \frac{\nu^2}{2}\xi^2 + \mathcal{O}(\xi^3), \quad \nu \equiv a\Delta t/\Delta x,$$

matching Lax-Wendroff at leading orders. Sympy-verified the three coefficients. Numerical sweep confirms $|g|^2 \leq 1$ for $\nu \in \{0.1, \dots, 1.0\}$; at $\nu = 1.05$ the amplification is $|g|^2 \approx 1.22 > 1$, confirming the CFL limit is **sharp**.

8.4 Truncation error (A7-consistency)

With the central-flux semi-discrete operator $\mathcal{L}[U]_j = -a(U_{j+1} - U_{j-1})/(2h)$ acting on smooth U , after two VL2 stages,

$$\mathbf{U}_j^{n+1} - U(x_j, t + \Delta t) = \mathcal{O}(\Delta t^3 + \Delta t \Delta x^2).$$

Leading error is a purely dispersive $U_{xxx} a(a^2 - 1)/6$ term, vanishing at $\nu = 1$ (exact for advection on-CFL). Sympy expands the predictor/corrector composition and confirms the residual has no $\mathcal{O}(\varepsilon^0 \dots \varepsilon^2)$ terms (with $\varepsilon = \Delta t = h$).

8.5 CFL constraint (1D)

$$\Delta t \leq C_{\text{CFL}} \frac{\Delta x}{\max_{\text{cells}}(|v_x| + c_f)}, \quad C_{\text{CFL}} \leq 1.$$

See §A8 for multidimensional generalisation and parabolic terms.

8.6 Implementation tips (for athena_mhd_solver.cu)

- Save $\mathbf{U}^n$ **before** the predictor so the corrector can rewind if a positivity check fails.
- The predictor need **not** compute the flux to full order — a 1st- order Godunov flux is enough; the corrector is where the 2nd-order accuracy is paid for. Stone+08 explicitly notes this economy.
- For CT, compute **face-centred** E_z at both predictor and corrector; average at the corners using Gardiner-Stone 2005 (§A5).

8.7 □ Verification checkpoints

- tests/test_athena_mhd_linear_wave_convergence.cu — three resolutions, fast wave, expect L^1 convergence slope in $[1.9, 2.1]$.
- tests/test_athena_mhd_vl2_stability.cu — CFL scan $\nu \in \{0.5, 0.9, 0.99\}$ stable; $\nu = 1.05$ must blow up.

9 A8. MHD CFL time-step constraints

sympy script: scripts/a8_mhd_cfl.py **verified:** FTCS diffusion amplification worst-case at $\xi = \pi$ gives $\sigma \leq 1/2$; c_f limits in three degenerate cases ($B_\perp = 0$, $B_x = 0$, $\mathbf{B} = 0$). **code checkpoints:** athena_mhd_kernels.cu::d_mhd_dt_reduction, athena_mhd_solver.cu::compute_dt.

9.1 Hyperbolic CFL (fast wave dominant)

Combining §A3 (7-wave eigensystem) with §A7 ($|\nu| \leq 1$), the unsplit multidimensional bound is

$$\Delta t_{\text{hyp}} \leq C_{\text{CFL}} \left/ \sum_{d=1}^D \max_{\text{cells}} \left(\frac{|v_d| + c_{f,d}}{\Delta x_d} \right) \right., \quad C_{\text{CFL}} \leq 1,$$

where the fast-magnetosonic speed $c_{f,d}$ in direction d is the positive root of the §A3 discriminant.

9.2 Parabolic CFL (Ohmic + ambipolar diffusion)

For the scalar diffusion $\partial_t U = \eta \partial_x^2 U$ with FTCS central space + forward Euler time, the amplification factor is $g(\xi) = 1 - 4\sigma \sin^2(\xi/2)$ with $\sigma = \eta \Delta t / \Delta x^2$. The worst case $\xi = \pi$ gives $g = 1 - 4\sigma$; stability $|g| \leq 1$ requires $\sigma \leq 1/2$ (sympy-verified).

Combining Ohmic + ambipolar as $\eta_{\text{eff}} = \eta_\Omega + \eta_{\text{AD}}$:

$$\Delta t_{\text{para}} \leq \frac{1}{2} \min_{\text{cells}} \frac{\Delta x^2}{\eta_{\Omega} + \eta_{\text{AD}}}.$$

9.3 Fast-speed behaviour at degenerate limits (A8-cf-limits)

Sympy-verified:

Limit	c_f^2
$B_{\perp} = 0$ (tangential-free)	$\max(c_{s_0}^2, c_{Ax}^2)$
$B_x = 0$ (perpendicular only)	$c_{s_0}^2 + c_{A\perp}^2$
$\mathbf{B} = 0$ (hydro)	$c_{s_0}^2$

The first two use numerical fall-back in sympy because `sp.Max` does not simplify against the nested radical; 30 random states confirm agreement to machine precision.

9.4 Combined rule

$$\Delta t = \min(\Delta t_{\text{hyp}}, \Delta t_{\text{para}}).$$

9.5 Kernel-form reduction

Implementation-wise, per cell:

$$\left(\frac{1}{\Delta t}\right)_{i,j,k} = \frac{|v_x| + c_f}{\Delta x} + \frac{|v_y| + c_f}{\Delta y} + \frac{|v_z| + c_f}{\Delta z} + 2 \frac{\eta_{\Omega} + \eta_{\text{AD}}}{\min(\Delta x, \Delta y, \Delta z)^2}.$$

Then Δt^{-1} is reduced (max) over the grid and the global Δt is $C_{\text{CFL}}/\Delta t_{\text{max}}^{-1}$.

9.6 Super-time-stepping (optional, future)

When η is large enough that $\Delta t_{\text{para}} \ll \Delta t_{\text{hyp}}$, the RKL2 super-time-stepping scheme (Meyer+12) advances the parabolic part with N sub-steps per hyperbolic step at $\mathcal{O}(N^2)$ CFL relaxation. Not included in the initial athena_mhd implementation; reserved for §C5 future extension when Suzuki turbulent heating is active.

9.7 □ Verification checkpoints

- `tests/test_athena_mhd_cfl_advection.cu` — advection of smooth wave at $C_{\text{CFL}} = 0.95$; no growth. At $C_{\text{CFL}} = 1.05$ the test **must** go unstable within 10 steps (positive control).
- `tests/test_athena_mhd_ohm_diffusion.cu` — pure Ohmic field-loop decay; Δt_{para} honored; analytic decay rate matched to $< 10^{-5}$.

10 A9. HLLD degenerate branches

sympy script: scripts/a9_hlld_degeneracy.py **verified:** on the Alfvén locus $(S_K - v_{xK})^2 = B_x^2/\rho_K$, numerator of B_{yK}^* vanishes and denominator reduces to $\rho_K(S_K - v_{xK})(v_{xK} - S_M)$; $D_{S_M} = 0$ when both sides of the Riemann interface are identical. **code checkpoints:** athena_mhd_kernels.cu::d_hlld_flux (branch dispatch).

10.1 Three degeneracies of the generic HLLD

The §A4 formulas B_{yK}^*, v_{yK}^* blow up in three cases. The kernel must dispatch on these **before** evaluating the generic formulas.

Branch	Condition	Fallback
D1	$B_x = 0$	HLLC-MHD (Li 2005), 3-wave fan
D2	$(S_K - v_{xK})^2 = B_x^2/\rho_K$	Upstream B_y, v_y, B_z, v_z unchanged
D3	$S_L \rightarrow S_R$ and sides identical	Pure HLL (2-wave average)

10.2 D1 — Zero longitudinal field ($B_x = 0$)

$$B_x = 0 \implies S_L^* = S_R^* = S_M, \mathbf{U}_K^* = \mathbf{U}_K^{**},$$

so the two-plateau star structure collapses to one plateau. The algorithm reduces to **HLLC-MHD** (Li 2005 Appendix A). Recommended implementation: detect $|B_x| < \varepsilon \cdot (c_{s_0} + c_{A\perp})\sqrt{\rho}$ and dispatch.

10.3 D2 — Alfvén locus removable singularity

The MK Eq. 44 denominator $\rho_K(S_K - v_{xK})(S_K - S_M) - B_x^2$ vanishes on the locus $(S_K - v_{xK})^2 = B_x^2/\rho_K$, where the upstream state already sits on the Alfvén characteristic.

Sympy-verified along the locus: - Numerator $\rho_K(S_K - v_{xK})^2 - B_x^2 \rightarrow 0$. - Denominator reduces to $\rho_K(S_K - v_{xK})(v_{xK} - S_M)$.

So both vanish, and B_{yK}^* is $0/0$ — *removable*. The physical content is that the transverse field does not jump across an Alfvén wave that coincides with the upstream state.

Kernel regularisation:

$$|\text{den}| < \epsilon\sqrt{\rho_K}|B_x| \implies B_{yK}^* \leftarrow B_{yK}, v_{yK}^* \leftarrow v_{yK} \text{ (and same for } z\text{)}.$$

with $\epsilon \sim 10^{-12}$ in double precision. Falling through to the upstream state is continuous with the generic formula.

10.4 D3 — $S_L \approx S_R$ (vacuum / strongly-aligned)

When $\|(S_R - v_{xR})\rho_R - (S_L - v_{xL})\rho_L\| < \epsilon$, the S_M denominator vanishes. Physically: the Riemann fan has collapsed ($L = R$), so there is no jump to resolve. Fall back to pure HLL:

$$\mathbf{F}_{\text{HLL}} = \frac{S_R \mathbf{F}_L - S_L \mathbf{F}_R + S_L S_R (\mathbf{U}_R - \mathbf{U}_L)}{S_R - S_L}.$$

Pure HLL is diffusive but unconditionally stable, a safe last-resort.

10.5 Implementation order (most to least specific)

```
if |D_SM| < ε:
    return HLL_flux()           # D3
if |B_x| < ε * (cs0 + cA1)*sqrt(ρ):
    return HLLC_MHD_flux()      # D1
compute generic star states (§A4)
if |den_K| < ε * sqrt(ρ_K) * |B_x|:  # D2
    set B_y*_K, B_z*_K, v_y*_K, v_z*_K
    to upstream values
return HLLD_flux(generic)
```

10.6 Why this matters

Without these dispatches, the Brio-Wu shock tube **will** produce NaN at $t \sim 0.05$ when a rarefaction fan grazes the Alfvén locus. Documented in Miyoshi-Kusano 2005 Sec. 3.4 and in Athena++ `src/hydro/rsolvers/mhd/hlld.cpp::HLLDTransport` branch logic.

10.7 □ Verification checkpoints

- `tests/test_athena_mhd_brio_wu.cu` — Brio-Wu, 512 cells, match Stone+08 Fig 28 to $L^1 < 2\%$.
- `tests/test_athena_mhd_degenerate_Bx0.cu` — synthetic state with $B_x = 10^{-14}$, must not produce NaN and must match pure-HLLC hydrodynamic flux to 10^{-10} .

11 A10. Powell 8-wave source vs Constrained Transport

sympy script: `scripts/a10_powell_vs_ct.py` **verified:** Powell source term $\mathbf{S}_p$ vanishes identically when $\nabla \cdot \mathbf{B} = 0$; since §A5 gives CT preservation at the discrete level, $\mathbf{S}_p \equiv 0$ for all n and no correction is needed. **code checkpoints:** no new code required — documented to prevent future “add a Powell source term for safety” PR.

11.1 The 8-wave system (Powell+99)

The 8-wave formulation augments ideal MHD with a divergence-cleaning wave and a **non-conservative** source term:

$$\partial_t \mathbf{U} + \partial_i \mathbf{F}_i(\mathbf{U}) = -(\nabla \cdot \mathbf{B}) \mathbf{S}_p(\mathbf{U}),$$

with

$$\mathbf{S}_p = \begin{pmatrix} 0 \\ \mathbf{B} \\ \mathbf{v} \\ \mathbf{B} \cdot \mathbf{v} \end{pmatrix}.$$

At the **continuous** level, $\nabla \cdot \mathbf{B} = 0$ exactly, so $\mathbf{S}_p \cdot 0 = 0$ — trivial.

11.2 The non-trivial claim

At the **discrete** level, a cell-centred or vertex-centred $\mathbf{B}$ storage generally has $(\nabla \cdot \mathbf{B})_{i,j} \neq 0$ at $\mathcal{O}(\Delta x)$, and the Powell source is a genuine correction. **But** on the Yee-staggered grid with CT (§A5), the discrete $(\nabla \cdot \mathbf{B})_{i,j}^n$ satisfies

$$(\nabla \cdot \mathbf{B})_{i,j}^n = (\nabla \cdot \mathbf{B})_{i,j}^0 \quad \forall n$$

(the §A5 telescoping identity). Provided initialisation seeds $\mathbf{B}^0$ from a vector potential or via one projection solve $(\nabla^2 \phi = \nabla \cdot \mathbf{B}_{\text{raw}}, \mathbf{B} \leftarrow \mathbf{B}_{\text{raw}} - \nabla \phi)$, $(\nabla \cdot \mathbf{B})^0 = 0$ and thus

$$\boxed{\mathbf{S}_p^n \equiv 0 \quad \forall n.}$$

11.3 GLM-MHD contrast (Dedner+02)

GLM adds an 8th field ψ and hyperbolic-parabolic cleaning:

$$\partial_t \psi + c_h^2 (\nabla \cdot \mathbf{B}) = -\alpha \psi, \quad \partial_t \mathbf{B} + \nabla \psi = \dots$$

GLM **does not** preserve $\nabla \cdot \mathbf{B} = 0$ exactly; it advects and damps the constraint violation at wave speed c_h . CT is *exact* by construction; GLM trades exactness for a grid that does not need face-centred field storage (ADER-DG codes etc.).

For `athena_mhd` we use CT. This section exists to lock in “no Powell source needed” as a *derived* result, not an assumption.

11.4 Implementation note

The kernel must:

1. Seed $\mathbf{B}^0$ from a vector potential $\mathbf{A}$ via $\mathbf{B} = \nabla \times \mathbf{A}$ at face centres. No projection needed; $\nabla \cdot \mathbf{B} \equiv 0$ by vector calculus.
2. Run the diagnostic `d_mhd_divB_check` after every time step to confirm $\max |\nabla \cdot \mathbf{B}| < 10 \epsilon_{\text{mach}}$. If this check ever fails, the bug is in Step 1 or in the CT kernel; not a reason to add a Powell source.

11.5 □ Verification checkpoints

- `tests/test_athena_mhd_field_loop.cu` — Gardiner-Stone 2005 field loop, 10 diagonal advections. Track $\max |\nabla \cdot \mathbf{B}|$ — must remain at machine precision.
- No test for the Powell source — it is **not** in the kernel and should never be added.

12 A11. Linear MHD wave initial conditions (Stone+08 Tab 1)

sympy script: `scripts/all_linear_wave_initial_conditions.py` **verified:** with $\rho_0 = 1, p_0 = 1/\gamma, \mathbf{B}_0 = (1, \sqrt{2}, 1/2), \gamma = 5/3$ the wave speeds are $c_f = 2, c_s = 1/2, c_{Ax} = 1, c_{s_0} = 1$, all matching Stone+08 Table 1; the discriminant identities $c_f^2 + c_s^2 = c_{s_0}^2 + c_A^2$ and $c_f^2 c_s^2 = c_{s_0}^2 c_{Ax}^2$ are sympy-verified. **code checkpoints:** `tests/test_athena_mhd_linear_wave_convergence.cu; tst/test_athena_mhd/test_linwave.py` (Python harness analogous to the existing `tst/test_ale2/test_linwave.py`).

12.1 Why this is the “gold-standard” convergence test

Linear wave advection has a known exact solution: after one period the perturbation returns to its initial shape, and any deviation is **purely numerical**. The L^1 norm of the difference, sampled at multiple resolutions, directly measures convergence slope and diagnoses accuracy loss of the scheme.

A scheme that claims 2nd order but gives slope 1.3 in the fast-wave test has a broken reconstruction, broken HLLD, or broken CT — and this test finds that before `athena_mhd` touches a shock tube.

12.2 Background (Stone+08 Table 1)

$$\rho_0 = 1, \quad p_0 = \frac{1}{\gamma}, \quad \mathbf{v}_0 = \mathbf{0}, \quad \mathbf{B}_0 = \left(1, \sqrt{2}, \frac{1}{2}\right), \quad \gamma = \frac{5}{3}.$$

Derived wave speeds (all sympy-verified):

$$c_f = 2, \quad c_{Ax} = 1, \quad c_s = \frac{1}{2}, \quad c_{s_0} = 1.$$

12.3 Perturbation

For mode k with eigenvector $\mathbf{r}_k$ (from §A3) and amplitude $A = 10^{-6}$:

$$\mathbf{W}(x, 0) = \mathbf{W}_0 + A \mathbf{r}_k \cos\left(\frac{2\pi x}{L}\right), \quad x \in [0, L].$$

12.4 Periods on an L -periodic box

$$T_f = L/2, \quad T_A = L, \quad T_s = 2L.$$

12.5 Entropy-mode exception

The entropy eigenvalue is $\lambda_{\text{ent}} = v_{0,x}$ which is 0 at our base state. To make the entropy-wave convergence test meaningful, **choose** $v_{0,x} = 1$ **for the entropy-mode case only**; then $T_{\text{ent}} = L$. Stone+08 does the same (Sec 6.1).

12.6 $\nabla \cdot \mathbf{B} = 0$ at IC (automatic)

In the 1D MHD system (§A3) B_x is a parameter, not an evolution variable; all seven right-eigenvectors live in $(\rho, v_x, v_y, v_z, B_y, B_z, p)$ and do not perturb B_x . For a plane wave along $\hat{x}$, $\nabla \cdot \delta \mathbf{B} = i k_x \delta B_x = 0$ automatically. **No extra projection step needed at IC.**

12.7 Expected error

For a VL2 + HLLD + CT code (§A7, §A4, §A5),

$$\varepsilon_{L^1}(k, \Delta x) = C_k A (\Delta x/L)^2,$$

with C_k an $\mathcal{O}(1)$ mode-dependent constant. Convergence slope measured across $\Delta x = L/64, L/128, L/256$ should fall in $[1.9, 2.1]$ for all 7 modes.

12.8 Practical test protocol

1. For each of the 7 modes:
 - Run the solver at resolutions $N \in \{64, 128, 256\}$.
 - Evolve for one period T_k .
 - Compute $\varepsilon_{L^1}(N) = \frac{1}{N} \sum_j |W_j^N - W_j^{\text{exact}}|$.
2. Fit slope: $\log \varepsilon \propto p \log \Delta x$.
3. Accept if $|p - 2| < 0.1$.

12.9 □ Verification checkpoint

tests/test_athena_mhd_linear_wave_convergence.cu — single test that cycles through 7 modes $\times$ 3 resolutions $\times$ 1 period; outputs a 21-row CSV; asserts slope $\in [1.9, 2.1]$ per mode.

12.10 Extension: 3D linear wave

A 3D linear wave IC is constructed identically, substituting $k_x \rightarrow \mathbf{k}$ and rotating the eigenvector by the same angle. Stone+08 §6.2 does the oblique test at $\mathbf{k} = (1, 2, 2)$. This stresses dimensional coupling. Reserved for post-MVP; not included in the initial test suite.

13 B1. Super-radial flux-tube reduction

sympy script: scripts/b1_flux_tube_geometry.py **verified:** flux conservation $AB_r = \text{const}$; MHSE $\partial_r p + \rho g = -B_r^2 \partial_r (\ln A)$; Kopp-Holzer $f(R_*) = 1$, $f(\infty) = f_{\text{max}}$. **code checkpoints:** tests/test_mhd_wind_hse_stationary.cu.

13.1 Geometry

$A(r) = r^2 f(r)$, $f(R_*) = 1$, $f(\infty) = f_{\text{max}}$. Kopp-Holzer form:

$$f(r) = \frac{f_{\text{max}} e^{(r-R_1)/\sigma} + f_1}{e^{(r-R_1)/\sigma} + 1}, \quad f_1 = 1 - (f_{\text{max}} - 1) e^{(R_*-R_1)/\sigma}.$$

13.2 1D flux-tube equations

$$\begin{aligned}\partial_t(\rho A) + \partial_r(\rho v_r A) &= 0, \\ \partial_t(\rho v_r A) + \partial_r[A(\rho v_r^2 + p_{\text{tot}} - B_r^2)] &= (p_{\text{tot}} - B_r^2)\partial_r A - \rho g A, \\ \partial_t(EA) + \partial_r[A((E + p_{\text{tot}})v_r - B_r(\mathbf{B} \cdot \mathbf{v}))] &= -\rho g v_r A + A Q.\end{aligned}$$

13.3 Flux conservation

$$A(r)B_r(r) = R_*^2 B_0 \Rightarrow B_r(r) = \frac{B_0}{f(r)} (R_*/r)^2.$$

13.4 MHSE

$$\partial_r p + \rho g = -B_r^2 \partial_r (\ln A).$$

IC gotcha: for a Suzuki-style wind run, atmospheres must be integrated from this MHSE, NOT plain HSE. Using HSE triggers a $\sim 10^{-2}$ transient and corrupts $\dot{M}$ measurements.

13.5 WKB amplitude (see B2)

$\delta v_\perp \propto (\rho v_A A)^{-1/2}$ (Poynting) or $(\rho v_A A)^{-1/4}$ (per-mode amplitude convention).

13.6 □ Verification

tests/test_mhd_wind_hse_stationary.cu — HSE atmosphere, 100 acoustic crossings, lock $\max |v_r|/c_s < 10^{-4}$.

14 B2. WKB wave action conservation for Alfvén waves

sympy script: scripts/b2_wave_action_wkb.py **verified:** dispersion $\omega^2 = v_A^2 k^2$; no-reflection limit under uniform $\rho_0, B_{r,0}$. **code checkpoints:** athena_mhd_kernels.cu::d_alfven_wave tests/test_mhd_wind_amplitude_scaling.cu.

14.1 Linearised transverse MHD

Around a static background $(\rho_0(r), B_{r,0}(r))$ with $v_{r,0} = 0$:

$$\partial_t v_\perp = \frac{B_{r,0}}{\rho_0} \partial_r B_\perp, \quad \partial_t B_\perp = B_{r,0} \partial_r v_\perp. \quad (\text{B2-linear})$$

WKB ansatz gives the Alfvén dispersion

$$\omega^2 = v_A^2 k^2, \quad v_A \equiv B_{r,0}/\sqrt{\rho_0}. \quad (\text{B2-disp})$$

Sympy verified that the determinant of the 2×2 linearised matrix equals $-\omega^2 + v_A^2 k^2$, giving the dispersion as the unique non-trivial mode.

14.2 Elsässer variables

$z_{\pm} \equiv v_{\perp} \mp B_{\perp}/\sqrt{\rho_0}$ decouple the outgoing and incoming Alfvén waves:

$$\partial_t z_{\pm} \pm v_A \partial_r z_{\pm} = S_{\text{refl}} z_{\mp}, \quad (\text{B2-Elsasser})$$

with a reflection source S_{refl} that depends on the background gradients. In a uniform $(\rho_0, B_{r,0})$, sympy verifies $S_{\text{refl}} = 0$ — no wave reflection in a uniform atmosphere (Alfvén waves are pure one-way in the uniform limit).

14.3 Wave action conservation

Outgoing z_+ in the absence of reflection conserves the wave action

$$\boxed{\rho_0 A |z_+|^2 / v_A = \text{const along characteristics.}} \quad (\text{B2-action})$$

For the amplitude-scaling law of a driver BC (relevant for Suzuki winds), **Poynting flux** $\rho_0 v_A A |\delta v_{\perp}|^2$ conservation gives

$$|\delta v_{\perp}| \propto (\rho_0 v_A A)^{-1/2}, \quad (\text{B2-Poynting})$$

while a **per-mode** amplitude normalisation gives the Jacques-1977 form

$$|\delta v_{\perp}| \propto (\rho_0 v_A A)^{-1/4}. \quad (\text{B2-Jacques})$$

The two differ because the former fixes the transmitted energy flux while the latter fixes the wave amplitude in Elsässer space. Suzuki wind codes use **the Poynting-flux convention** at the photospheric driver (amplitude $\langle \delta v_{\perp} \rangle \approx 1.25$ km/s in Shimizu+22) — make sure the kernel matches.

14.4 □ Verification

tests/test_mhd_wind_amplitude_scaling.cu — linearised Alfvén wave from a fixed BC driver. Measure $\delta v_{\perp}^{\text{rms}}(r)$ at $r = 5, 10, 20 R_*$; lock deviation from $(\rho v_A A)^{-1/2}$ at $< 5\%$.

15 B3. Parker critical point on a super-radial flux tube

sympy script: scripts/b3_parker_critical_point.py **verified:** Parker wind equation derivation; classical spherical limit $r_c = GM_*/(2c_s^2)$. **code checkpoints:** tests/test_mhd_parker_wind.cu.

15.1 Parker wind equation

Steady isothermal flow on a super-radial tube ($A(r) = r^2 f(r)$) with $p = c_s^2 \rho$ yields, after eliminating ρ via mass conservation:

$$\boxed{\left(v - \frac{c_s^2}{v}\right) \frac{dv}{dr} = c_s^2 \frac{d \ln A}{dr} - \frac{GM_*}{r^2}.} \quad (\text{B3-Parker})$$

Sympy derived from mass + momentum + EOS.

15.2 Critical (sonic) point

At $v = c_s$ the LHS vanishes, forcing

$$c_s^2 \left(\frac{2}{r_c} + \frac{d \ln f}{dr} \Big|_{r_c} \right) = \frac{GM_*}{r_c^2}. \quad (\text{B3-critical})$$

Spherical limit $f \equiv 1$: $r_c = GM_*/(2c_s^2)$ (classical Parker radius). Sympy verifies this by solving (B3-critical) analytically.

15.3 Asymptotic velocity

Far from r_c (spherical limit), $v(r) \sim c_s \sqrt{4 \ln(r/r_c) + \text{const}}$ — logarithmic growth, characteristic of the Parker isothermal wind.

15.4 □ Verification

tests/test_mhd_parker_wind.cu — isothermal, no magnetic driver, set $T = 2 \times 10^6$ K, $M_* = 1M_\odot$, measure Mach number crossing at $r = r_c$. Lock $|M(r_c) - 1| < 10^{-3}$.

16 B4. Well-balanced MHSE at the operator level

sympy script: scripts/b4_well_balanced_mhse.py **verified:** $F_{\text{wb}}(\mathbf{U}_{\text{hse}}) \equiv 0$ by construction; linear-wave Jacobian preserved, $\partial_{\mathbf{U}} F_{\text{wb}} = \partial_{\mathbf{U}} R$ at $\mathbf{U}_{\text{hse}}$ — perturbation dynamics unchanged. **code checkpoints:** athena_mhd_solver.cu::compute_residual_wb, athena_mhd_solver.cu::snapshot_hse_if_needed, tests/test_athena_mhd_wind_hse_stationary.cu.

16.1 The problem

Direct application of the discrete residual $R(\mathbf{U}) = -\partial_i \mathbf{F}_i + \mathbf{S}$ to a piecewise-constant MHSE atmosphere leaves a truncation residual of order $\mathcal{O}(\Delta r^2) \rho g$. For a Suzuki-style wind run this residual drives a $|\delta v_r|/c_s \sim 10^{-2}$ transient that corrupts the wind-mass-loss diagnostic for hundreds of crossing times.

16.2 Well-balanced residual (Bermúdez-Vázquez 1994 / Botta+04)

$$\boxed{F_{\text{wb}}(\mathbf{U}) \equiv R(\mathbf{U}) - R(\mathbf{U}_{\text{hse}})}.$$

By construction $F_{\text{wb}}(\mathbf{U}_{\text{hse}}) = 0$ (sympy-verified trivially). So an atmosphere at MHSE stays at MHSE to machine precision for arbitrarily long time, regardless of reconstruction order or Riemann solver.

16.3 Preservation of linear-wave dynamics

A small perturbation $\delta \mathbf{U}$ around the MHSE state:

$$F_{\text{wb}}(\mathbf{U}_{\text{hse}} + \delta \mathbf{U}) = \partial_{\mathbf{U}} R|_{\mathbf{U}_{\text{hse}}} \cdot \delta \mathbf{U} + \mathcal{O}(\delta \mathbf{U}^2).$$

Sympy-verified that the Jacobian of F_{wb} at $\mathbf{U}_{\text{hse}}$ **equals** the Jacobian of R at the same state. Thus linear MHD waves (Alfvén, magnetosonic) propagate through F_{wb} identically to R — no spurious damping or dispersion is introduced by the well-balancing subtraction.

16.4 What MHSE means in a super-radial flux tube

Reprise from §B1:

$$\partial_r p + \rho g + B_r^2 \partial_r (\ln A) = 0.$$

Note the **last term** is critical. A naïve WB that only cancels gravity (the hydrodynamic well-balancing) will leave an $\mathcal{O}(B_r^2)$ residual. The correction for the super-radial tube must include the area-divergence term.

16.5 Practical recipe

1. **Snapshot MHSE on the grid.** Integrate the ODE $\partial_r p_{\text{hse}} = -\rho_{\text{hse}} g - B_r^2 \partial_r (\ln A)$ using the given $(\rho_{\text{hse}}, B_r, A)$ profiles.
2. **Compute $R(\mathbf{U}_{\text{hse}})$ once** at startup, store per cell. (Assumes $\mathbf{U}_{\text{hse}}$ is time-independent — true for stationary background.)
3. **Evolve** $\partial_t \mathbf{U} = R(\mathbf{U}) - R(\mathbf{U}_{\text{hse}})$.
4. **Initial condition:** seed $\mathbf{U}^0 = \mathbf{U}_{\text{hse}}$. Initial RHS identically zero $\rightarrow$ no transient.

16.6 Failure modes observed elsewhere

- radial1d without this correction: Newton iteration looks converged ($|F| < 10^{-9}$) but the HSE slowly drifts at machine precision $\times$ resolution. Seen in pre-MS KH attempts: HSE-Newton is stable but cannot initiate KH because $F_v \equiv 0$ at MHSE.
- cart_ale2 with WB: integration-grade HSE stability for 10^4 crossings.

16.7 Snapshot invalidation

If $(\rho_{\text{hse}}, B_r, A)$ profiles change (e.g., user-driven parameter sweep), re-snapshot. The kernel can detect staleness via a hash of the profile arrays.

16.8 □ Verification checkpoints

- tests/test_athena_mhd_wind_hse_stationary.cu — MHSE atmosphere, 10^4 acoustic crossings, assert $\max |v_r|/c_s < 10^{-8}$ (with WB on) vs $\sim 10^{-2}$ (with WB off). The “off” case is a **positive control**: if turning off WB doesn’t produce a transient, the WB machinery is a no-op and should be audited.

17 C1. Ohmic (resistive) dissipation

sympy script: scripts/cl_ohmic_dissipation.py **verified:** $\nabla \times (\nabla \times B) = \nabla(\nabla \cdot B) - \nabla^2 B$ (all 3 components); $\nabla \times (\eta_O J) = \eta_O \nabla \times J + (\nabla \eta_O) \times J$. **code checkpoints:** athena_mhd_kernels.cu::d_ohmic_dissipation, tests/test_mhd_ohmic_diffusion.cu.

17.1 Ohm's law with finite conductivity

$$\mathbf{E} = -\mathbf{v} \times \mathbf{B} + \eta_O \mathbf{J}, \quad \mathbf{J} = \nabla \times \mathbf{B}. \quad (\text{C1-Ohm})$$

17.2 Induction equation

Constant η_O :

$$\partial_t \mathbf{B} = \nabla \times (\mathbf{v} \times \mathbf{B}) + \eta_O \nabla^2 \mathbf{B}. \quad (\text{C1-induction-const})$$

Variable $\eta_O(\mathbf{r})$:

$$\partial_t \mathbf{B} = \nabla \times (\mathbf{v} \times \mathbf{B}) + \eta_O \nabla^2 \mathbf{B} - (\nabla \eta_O) \times \mathbf{J}. \quad (\text{C1-induction-var})$$

Sympy verified via the vector identity $\nabla \times (\eta_O \mathbf{J}) = \eta_O \nabla \times \mathbf{J} + (\nabla \eta_O) \times \mathbf{J}$.

17.3 Joule heating

$$Q_{\text{Ohm}} = \eta_O |\mathbf{J}|^2 \geq 0. \quad (\text{C1-Q})$$

Manifestly non-negative; enters internal-energy source.

17.4 CFL for explicit diffusion

$$\Delta t \leq \frac{(\Delta x)^2}{2\eta_O} \text{ (1D)}, \frac{(\Delta x)^2}{4\eta_O} \text{ (2D)}, \frac{(\Delta x)^2}{6\eta_O} \text{ (3D)}. \quad (\text{C1-CFL})$$

At $r \approx 1000$ km in Matsuoka+24, $\eta_O \sim 10^{12}$ cm²/s and $\Delta x \sim 1$ km, so $\Delta t_{\text{Ohm}} \sim 10^{-4}$ s — much smaller than hydro CFL. In practice, Matsuoka+24 use **super-time-stepping** (Alexiades-Amiez-Gremaud 1996) to accelerate the explicit Ohmic update.

17.5 □ Verification

tests/test_mhd_ohmic_diffusion.cu — sinusoidal B_y perturbation, $\eta_O = \text{const}$. Lock $L^2(B_y)$ exponential decay rate matches analytical $e^{-\eta_O k^2 t}$ to $< 1\%$ at $N = 128$.

18 C2. Ambipolar diffusion (ion-neutral drift)

sympy script: scripts/c2_ambipolar_dissipation.py **verified:** $\mathbf{J}_\perp \cdot \hat{\mathbf{B}} = 0$; $\mathbf{J}_\parallel \times \mathbf{B} = 0$; $(\mathbf{J} \times \hat{\mathbf{B}}) \times \hat{\mathbf{B}} = -\mathbf{J}_\perp$; $(\mathbf{J} \times \mathbf{B}) \times \mathbf{B}/|\mathbf{B}|^2 = (\mathbf{J} \times \hat{\mathbf{B}}) \times \hat{\mathbf{B}}$; $Q_{\text{amb}} = \eta_A |\mathbf{J}_\perp|^2 \geq 0$. **code checkpoints:** athena_mhd_kernels.cu::d_ambipolar_flux, tests/test_mhd_ambipolar_bmin.cu.

18.1 Motivation

In a partially-ionised plasma, **neutrals** do not couple directly to $\mathbf{B}$; they feel the Lorentz force only via ion-neutral collisions. When the collision rate ν_{in} is slower than MHD timescales, ions drift relative to neutrals at

$$\mathbf{v}_{\text{drift}} = \mathbf{J} \times \mathbf{B} / (\rho_i \rho_n \nu_{in}),$$

yielding the non-ideal electric field

$$\mathbf{E}_{\text{amb}} = \eta_A (\mathbf{J} \times \hat{\mathbf{B}}) \times \hat{\mathbf{B}} = -\eta_A \mathbf{J}_{\perp}, \quad (\text{C2-Eamb})$$

with $\eta_A \equiv |\mathbf{B}|^2 / (\rho_i \rho_n \gamma_{in})$.

18.2 Parallel / perpendicular current selectivity

Critical property: only $\mathbf{J}_{\perp}$ is dissipated by ambipolar; the parallel current $\mathbf{J}_{\parallel}$ flows freely along $\hat{\mathbf{B}}$:

$$\mathbf{J}_{\parallel} \times \mathbf{B} = \mathbf{0}. \quad (\text{C2-selective})$$

18.3 Kernel-friendly form

For implementation, the non-unit-vector form is preferred:

$$\boxed{\mathbf{E}_{\text{amb}} = \eta_A \frac{(\mathbf{J} \times \mathbf{B}) \times \mathbf{B}}{|\mathbf{B}|^2}}, \quad (\text{C2-altform})$$

avoids dividing by $|\mathbf{B}|$ in the normalisation step — matters for low- $|\mathbf{B}|$ robustness (chromosphere bottoms).

18.4 Induction equation

$$\partial_t \mathbf{B} = \nabla \times (\mathbf{v} \times \mathbf{B}) + \nabla \times \left[\eta_A \frac{(\mathbf{J} \times \mathbf{B}) \times \mathbf{B}}{|\mathbf{B}|^2} \right]. \quad (\text{C2-induction})$$

Non-linear in $\mathbf{B}$ — key difference from Ohmic.

18.5 Ambipolar heating

$$\boxed{Q_{\text{amb}} = \eta_A |\mathbf{J}_{\perp}|^2 = \eta_A \frac{|\mathbf{J}|^2 |\mathbf{B}|^2 - (\mathbf{J} \cdot \mathbf{B})^2}{|\mathbf{B}|^2} \geq 0.} \quad (\text{C2-Q})$$

18.6 CFL

$$\Delta t \leq \frac{(\Delta x)^2}{2\eta_A |\mathbf{B}|^2 / \rho} \quad (\text{perdirection}). \quad (\text{C2-CFL})$$

Scales with $|\mathbf{B}|^2$ — tight CFL in strong-field regions.

18.7 □ Verification

tests/test_mhd_ambipolar_bmin.cu — B_x uniform, sinusoidal B_y with $\eta_A = \text{const}$. Lock decay rate against analytical; also verify that when $B_y \parallel B_x$ (i.e., $\mathbf{J} \parallel \mathbf{B}$), $Q_{\text{amb}} = 0$ and B_y does not decay. Catches any accidental scalar-diffusion substitution for the tensor form.

19 C3. Total-energy equation with non-ideal MHD

sympy script: `scripts/c3_resistive_energy.py` **verified:** ideal Poynting flux identity; $Q_{\text{Ohm}} = \eta_O |J|^2$; $Q_{\text{amb}} = \eta_A |J_\perp|^2$ (via Lagrange's identity). **code checkpoints:** `athena_mhd_kernels.cu::d_total_energy_flux`.

19.1 Total-energy equation

With both Ohmic + ambipolar, total energy is still conserved; only the Poynting flux gets a non-ideal addition:

$$\partial_t E + \nabla \cdot [(E + p^*)\mathbf{v} - \mathbf{B}(\mathbf{v} \cdot \mathbf{B}) + \mathbf{E}_{\text{ni}} \times \mathbf{B}] = 0, \quad (\text{C3-energy})$$

with $\mathbf{E}_{\text{ni}} = \eta_O \mathbf{J} + \eta_A (\mathbf{J} \times \mathbf{B}) \times \mathbf{B} / |\mathbf{B}|^2$.

19.2 Internal-energy source

The non-ideal dissipation becomes internal-energy heating:

$$\partial_t (\rho e) + \nabla \cdot (\rho e \mathbf{v}) = -p \nabla \cdot \mathbf{v} + Q_{\text{Ohm}} + Q_{\text{amb}}, \quad (\text{C3-internal})$$

$$Q_{\text{Ohm}} = \eta_O |\mathbf{J}|^2, \quad Q_{\text{amb}} = \eta_A |\mathbf{J}_\perp|^2 = \eta_A \frac{|\mathbf{J}|^2 |\mathbf{B}|^2 - (\mathbf{J} \cdot \mathbf{B})^2}{|\mathbf{B}|^2}.$$

Sympy verified the Lagrange identity $|\mathbf{J} \times \mathbf{B}|^2 = |\mathbf{J}|^2 |\mathbf{B}|^2 - (\mathbf{J} \cdot \mathbf{B})^2$ which maps between the two Q_{amb} forms.

19.3 Sign convention

$Q = -\mathbf{E}_{\text{ni}} \cdot \mathbf{J}$ is the rate at which EM energy is dissipated **into the fluid's internal energy** — positive for dissipative $\mathbf{E}_{\text{ni}}$. Both Ohmic and ambipolar give $Q > 0$.

19.4 Kernel structure

- **Total energy:** evolved via conservative flux (C3-energy). Non-ideal Poynting flux $\mathbf{E}_{\text{ni}} \times \mathbf{B}$ added to the hydro energy flux.
- **No direct Q kernel needed:** internal-energy source is implicitly accounted for by the flux divergence of the non-ideal Poynting term. Explicit computation of Q only needed for diagnostics.

19.5 □ Verification

`tests/test_mhd_ohmic_energy_conservation.cu` — uniform $\mathbf{B}_0$ with sinusoidal B_y perturbation + $\square_O = \square_{\text{const}}$. Measure total energy drift over 100 decay timescales. Lock $|\Delta E / E_0| < 10^{-8}$ — **any leak means the non-ideal Poynting flux term is missing from the energy flux**.

20 C4. Saha closure for η_O and η_A

sympy script: `scripts/c4_saha_ionization_closure.py` **verified:** Saha low-T limit $x_e \rightarrow 0$; high-T limit $x_e \rightarrow 1$. **code checkpoints:** `athena_mhd_kernels.cu::d_saha_eta, tests/test_mhd_saha_table.cu`.

20.1 Saha equation (pure hydrogen, LTE)

$$\frac{x_e^2}{1 - x_e} = \frac{1}{n_H} \left(\frac{m_e k_B T}{2\pi\hbar^2} \right)^{3/2} e^{-\chi_H/k_B T}, \quad (\text{C4-Saha})$$

with $\chi_H = 13.6$ eV and $x_e = n_e/(n_e + n_H)$. **Sympy verified** $x_e \rightarrow 0$ as $T \rightarrow 0$ and $x_e \rightarrow 1$ as $T \rightarrow \infty$.

20.2 Diffusivity closures (Draine 1983 / Choi+09)

Ohmic (electron-neutral collisions):

$$\eta_O \approx 234 \sqrt{T/10^4 \text{ K}} x_e^{-1} \text{ cm}^2/\text{s}. \quad (\text{C4-etaO})$$

Ambipolar (ion-neutral drift):

$$\eta_A = \frac{|\mathbf{B}|^2}{\rho_i \rho_n \gamma_{in}}, \quad \gamma_{in} \approx 3.5 \times 10^{13} \text{ cm}^3/\text{g/s}. \quad (\text{C4-etaA})$$

These are the forms used in Suzuki+25 (RGB winds) and Matsuoka+24 (solar chromosphere).

20.3 Weakly-ionised limit ($x_e \ll 1$)

$$\eta_O \sim \frac{234 \sqrt{T/10^4}}{x_e}, \quad \eta_A \sim \frac{|\mathbf{B}|^2}{x_e \rho^2 \gamma_{in}}.$$

Both diverge as $x_e \rightarrow 0$ — non-ideal effects are maximal at the chromosphere base where most hydrogen is neutral. This is why the RGB winds in Suzuki+25 get their $15\times \dot{M}$ suppression from ambipolar: it shuts down high-frequency Alfvén wave transmission at the chromospheric floor.

20.4 Magnetic Reynolds numbers

$$R_m^{\text{Ohm}} = \frac{Lv_A}{\eta_O}, \quad R_m^{\text{amb}} = \frac{Lv_A \rho}{\eta_A |\mathbf{B}|^2}.$$

Ideal MHD requires $R_m \gg 1$. Matsuoka+24 reports $R_m \sim 1 - 10$ at $r \approx 1000$ km in the solar chromosphere — non-ideal MHD kicks in below this altitude.

20.5 Kernel implementation

Pre-tabulate (η_O, η_A) as functions of $(\rho, T, |\mathbf{B}|)$ on a 3D table to avoid evaluating the Saha equation every timestep. Suzuki+25 uses a $128 \times 128 \times 64$ log-spaced table; cost is negligible once built.

20.6 □ Verification

tests/test_mhd_saha_table.cu — verify tabulated (η_O, η_A) match Draine 1983 Table 2 to $< 1\%$ at five reference (T, ρ) pairs: $(10^4, 10^{-9})$, $(6000, 10^{-7})$, $(4500, 10^{-8})$, $(3000, 10^{-11})$, $(10^5, 10^{-12})$.

21 C5. Sub-grid turbulent heating closure (Suzuki-Inutsuka 2005)

sympy script: scripts/c5_suzuki_turbulent_heating.py **verified:** positivity $\varepsilon_{\text{turb}} > 0$; pure-outward limit $|z^-| \rightarrow 0$ gives no cascade ($\varepsilon_{\text{turb}}^{\text{SY}} \rightarrow 0$); timescale bound $\Delta t \leq \lambda_{\text{cor}}/(2c_d|\delta v_\perp|)$. **code checkpoints:** athena_mhd_kernels.cu::d_turbulent_heating_source, tests/test_athena_mhd_turbulent_heating_positivity.cu.

21.1 Why this closure exists

The 1D super-radial flux-tube code (§B1) cannot resolve the 3D Alfvén wave turbulence that transfers outward-going wave energy to heat in the corona. Suzuki-Inutsuka 2005 (SI05) added a phenomenological sub-grid heating term that dissipates transverse wave energy at a prescribed cascade rate. This is the crucial ingredient without which the 1D wind models severely **under-estimate** coronal temperature.

21.2 Suzuki-Inutsuka 2005 form

$$\varepsilon_{\text{turb}}^{\text{SI}} = c_d \rho \frac{|\delta v_\perp|^3}{\lambda_{\text{cor}}}, \quad c_d \approx 0.1, \quad \lambda_{\text{cor}} \sim 10^7 \text{ cm.}$$

Dimensions: $[\rho][\delta v]^3/[\lambda] = \text{erg/cm}^3/\text{s}$ □.

21.3 Shoda-Yokoyama 2016 (Elsässer) form

$$\varepsilon_{\text{turb}}^{\text{SY}} = \frac{\rho}{2\lambda_{\text{cor}}} (|\mathbf{z}^+|^2|\mathbf{z}^-| + |\mathbf{z}^-|^2|\mathbf{z}^+|).$$

This form makes the **non-linear coupling** between counter-propagating modes explicit. Limit $|z^-| \rightarrow 0$ gives $\varepsilon_{\text{turb}}^{\text{SY}} \rightarrow 0$ (sympy-verified): no cascade without cross-polarisation — an essential physical consistency check.

21.4 Energy-equation coupling

$$\partial_t E + \nabla \cdot [(E + P^*)\mathbf{v} - \mathbf{B}(\mathbf{B} \cdot \mathbf{v})] = \varepsilon_{\text{turb}} \geq 0.$$

Positivity. Sympy-verified $\varepsilon_{\text{turb}} > 0$ for positive $\rho, \delta v_\perp, \lambda_{\text{cor}}, c_d$. The kernel must not inadvertently create a negative-definite heating source — any code review must check this.

21.5 Timescale bound (explicit-source CFL)

Per-step heating must not exceed the available wave KE:

$$\varepsilon_{\text{turb}} \cdot \Delta t \leq \frac{1}{2} \rho |\delta v_{\perp}|^2 \iff \Delta t \leq \frac{\lambda_{\text{cor}}}{2 c_d |\delta v_{\perp}|}.$$

In practice, for solar wind parameters ($|\delta v_{\perp}| \sim 30 \text{ km/s}$, $\lambda_{\text{cor}} \sim 10^8 \text{ cm}$), this bound is well above the hyperbolic CFL; no concern.

21.6 Shimizu+22 scaling

$$\lambda_{\text{cor}}(r) = \lambda_0 \sqrt{A(r)/A(R_*)}, \quad \lambda_0 \approx 10^7 \text{ cm}$$

(i.e., correlation length scales with the tube area because energy-containing eddies fill the tube cross-section). Replication of Shimizu+22 figures requires this scaling **and** $c_d = 0.1$ exactly — the paper’s calibration.

21.7 Extraction of $\delta v_{\perp}$

$\delta v_{\perp}$ is defined in the **Lagrangian / comoving frame**:

$$|\delta v_{\perp}|^2 \equiv \langle (v_y - \bar{v}_y)^2 + (v_z - \bar{v}_z)^2 \rangle,$$

with $\bar{v}$ a running time-average. In a 1D tube code with no transverse degree of freedom, this can instead be derived from the **Alfvén wave amplitude**

$$|\delta v_{\perp}|^2 = |z^+|^2 + |z^-|^2 - 2\mathbf{z}^+ \cdot \mathbf{z}^-$$

using the Elsässer variables tracked directly from §B2.

21.8 □ Verification checkpoints

- tests/test_athena_mhd_turbulent_heating_positivity.cu — random initial $(\rho, \delta v, \lambda)$ states, assert $\varepsilon_{\text{turb}} \geq 0$ across 100 samples; no NaN.
- tests/test_athena_mhd_suzuki_wind.cu — full replication of SI05 Fig. 2 mass-loss rate $\dot{M} \sim 2 \times 10^{-14} M_{\odot}/\text{yr}$ within $\pm 10\%$ (calibration tolerance per paper’s own sensitivity to c_d). Requires well-balanced MHSE (§B4).

22 D1. Ideal MHD in cylindrical (R, ϕ, z) coordinates

sympy script: scripts/d1_cylindrical_mhd.py **verified:** displays cylindrical divergence, curl, and axisymmetric reductions. No new nontrivial identities beyond Part A; this section is the reference for D2 and D3. **code checkpoints:** athena_mhd_cylindrical_solver.cu (future).

22.1 Cylindrical operators

$$\nabla \cdot \mathbf{v} = \frac{1}{R} \partial_R (R v_R) + \frac{1}{R} \partial_\phi v_\phi + \partial_z v_z.$$

$$(\nabla \times \mathbf{B})_R = \frac{1}{R} \partial_\phi B_z - \partial_z B_\phi, \quad (\nabla \times \mathbf{B})_\phi = \partial_z B_R - \partial_R B_z, \quad (\nabla \times \mathbf{B})_z = \frac{1}{R} [\partial_R (R B_\phi) - \partial_\phi B_R].$$

22.2 Radial momentum (axisymmetric)

$$\partial_t(\rho v_R) + \frac{1}{R} \partial_R (R \rho v_R^2) + \partial_z(\rho v_R v_z) = \boxed{\frac{\rho v_\phi^2}{R}} - \partial_R p - \rho \partial_R \Phi_{\text{grav}}. \quad (\text{D1-mom-R})$$

The $\rho v_\phi^2 / R$ term is the **Christoffel-symbol / centrifugal source**. It is what keeps a rotating disk in equilibrium against gravity. In a kernel, it is the most common source of subtle bugs — easy to drop when porting Cartesian kernels.

22.3 Axisymmetric radial-flux conservation

$\partial_\phi = \partial_z = 0$ with $\mathbf{B} = B_R \hat{R}$ forces $\nabla \cdot \mathbf{B} = (1/R) \partial_R (R B_R) = 0$, so $R \cdot B_R(R) = \text{const}$ — the cylindrical analog of the spherical $r^2 B_r$ conservation.

22.4 □ Verification

tests/test_mhd_cyl_centrifugal.cu — rotating equilibrium disk with $v_\phi = \sqrt{GM/R}$. Lock $\max |v_R|/c_s < 10^{-4}$ over 10 rotations. Failure means the centrifugal source term is missing or mis-signed.

23 D2. Shearing-sheet approximation and shearing-periodic BC

sympy script: scripts/d2_shearing_sheet_bc.py **verified:** shearing-sheet background is steady state (Coriolis + tidal cancel on v_ϕ^{bg}); shear wrap-around time $t_{\text{shear}} = L_y / (q \Omega_0 L_x)$; effective potential is anti-confining. **code checkpoints:** athena_mhd_shearingbox_bc.cu, tests/test_mhd_shearingbox_static.cu.

23.1 Hill expansion

Local Cartesian patch centred at (R_0, Ω_0) , co-rotating with the disk. Linearised rotational velocity:

$$\boxed{v_\phi^{\text{bg}}(x) = -q \Omega_0 x, \quad q \equiv -\frac{d \ln \Omega}{d \ln R}}. \quad (\text{D2-shear})$$

$q = 3/2$ for Keplerian disks, $q = 2$ for rigid rotation.

23.2 Frame-rotation sources

Coriolis:

$$\mathbf{F}_{\text{Cor}} = -2\Omega_0 \hat{z} \times \mathbf{u} = 2\Omega_0(u_y \hat{x} - u_x \hat{y}).$$

Tidal (Hill):

$$\mathbf{F}_{\text{tidal}} = 2q\Omega_0^2 x \hat{x} = -\nabla \Phi_{\text{eff}}, \quad \Phi_{\text{eff}} = -q\Omega_0^2 x^2.$$

Sympy verified that on the background $\mathbf{u}_{\text{bg}} = (0, -q\Omega_0 x, 0)$, $F_{\text{Cor},x} + F_{\text{tidal},x} = 0$. The shearing-sheet background is an exact steady state of the force-balance.

23.3 Shearing-periodic BC

$$\boxed{\text{field}(x = L_x, y, z, t) = \text{field}(x = 0, y - q\Omega_0 L_x t, z, t).} \quad (\text{D2-BC})$$

The y -shift $\Delta y(t) = q\Omega_0 L_x t$ grows linearly in time. After $t_{\text{shear}} = L_y/(q\Omega_0 L_x)$ the shift wraps to L_y and the BC becomes pure periodic again.

Under periodic- y convention, the BC is always well-defined: take $\Delta y \bmod L_y$. **Sympy verified** t_{shear} .

23.4 Effective potential is anti-confining

$$\frac{d^2 \Phi_{\text{eff}}}{dx^2} = -2q\Omega_0^2 < 0,$$

so the tidal potential alone is unstable — **the Coriolis force is what stabilises the shearing sheet**. Sympy-verified; documented to flag that dropping the Coriolis kernel would immediately disperse the disk.

23.5 □ Verification

tests/test_mhd_shearingbox_static.cu — initial condition $\mathbf{u} = (0, -q\Omega_0 x, 0)$, purely hydro, no perturbation. Lock max perturbation energy $< 10^{-6} \times E_0$ over $10/\Omega_0$. Any drift means Coriolis or tidal coupling is mis-implemented.

24 D3. MRI stress decomposition $\alpha_{\text{SS}} = \alpha_R + \alpha_M$

sympy script: scripts/d3_mri_stress_decomposition.py **verified:** Maxwell stress antisymmetric under $B_R \rightarrow -B_R$, invariant under $(B_R, B_\phi) \rightarrow (-B_R, -B_\phi)$; Reynolds stress antisymmetric under $v_R \rightarrow -v_R$; MRI dispersion vanishes at Balbus-Hawley extremum ($k_*^2 v_A^2 = 15/16 \Omega^2$, $\gamma_{\text{max}} = 3/4 \Omega$). **code checkpoints:** tests/test_mhd_mri_growth_rate.cu, tests/test_mhd_mri_stress_decomp.cu.

24.1 Shakura-Sunyaev stress

$$T_{R\phi} = \underbrace{\rho v_R \delta v_\phi}_{\text{Reynolds}} + \underbrace{(-B_R B_\phi)}_{\text{Maxwell}}. \quad (\text{D3-stress})$$

$$\alpha_{\text{SS}} = \frac{\langle T_{R\phi} \rangle}{\langle p \rangle} = \alpha_R + \alpha_M.$$

24.2 Suzuki 2023 sign-quadrant decomposition

Maxwell stress is: - **antisymmetric** under single flip $B_R \rightarrow -B_R$: $M_{R\phi} \rightarrow -M_{R\phi}$. - **invariant** under simultaneous flip $(B_R, B_\phi) \rightarrow (-B_R, -B_\phi)$.

This pair of symmetries is what makes the Suzuki 2305.12112 “triangle diagnostic” meaningful: the four sign quadrants $(\pm B_R, \pm B_\phi)$ carry physically distinct stress contributions. The cylindrical vs Cartesian sign-flip result in that paper (headline $[\phi \Rightarrow_R \phi]$ arrow +6.42 vs -1.05) is a direct consequence of this symmetry structure.

24.3 Balbus-Hawley MRI dispersion

For vertical $\mathbf{B}_0 = B_0 \hat{z}$ and Keplerian $q = 3/2$ ($\kappa^2 = \Omega^2$):

$$\omega^4 - \omega^2(\kappa^2 + 2k^2 v_A^2) + k^2 v_A^2 (k^2 v_A^2 + \kappa^2 - 4\Omega^2) = 0. \quad (\text{D3-BH-disp})$$

Instability requires $k^2 v_A^2 < 3\Omega^2$. The most-unstable mode:

$$\boxed{k_*^2 v_A^2 = \frac{15}{16}\Omega^2, \quad \gamma_{\max} = \frac{3}{4}\Omega.} \quad (\text{D3-MRI-max})$$

Sympy verified by direct substitution into the dispersion polynomial. This gives the MRI timescale $\sim 4/(3\Omega) \approx 1/3$ of an orbital period.

24.4 Parseval for Maxwell

$$\frac{1}{V} \int B_R(\mathbf{x}) B_\phi(\mathbf{x}) d^3x = \sum_{\mathbf{k}} \text{Re}[\hat{B}_R(\mathbf{k}) \hat{B}_\phi^*(\mathbf{k})]$$

Allows FFT-based computation of α_M at arbitrary resolution.

24.5 $\square$ Verification

tests/test_mhd_mri_growth_rate.cu — seed a linear B_y perturbation at k_* with amplitude $10^{-6} B_0$; lock measured growth rate matches $(3/4)\Omega$ to $< 1\%$ over $5/\Omega$.

tests/test_mhd_mri_stress_decomp.cu — in nonlinear saturated state, lock $\alpha_M/\alpha_R \sim 3 - 5$ (Stone+96, Suzuki+23 Cartesian baseline). Suzuki+23 reports $\alpha_M \approx 0.072$, $\alpha_R \approx 0.016$ for Cartesian Keplerian baseline.